

Direct Posterior Esthetic Restorations

Thomas J. Hilton
James C. Broome

The use of resin composite as a material for restoring posterior teeth has continued to increase. Patients are attracted to a restoration that matches the color of natural teeth.[1] Resin composite meets this demand and has become the most frequently used esthetic restorative material in dentistry.[2,3] In addition, resin composites avoid the concerns over the use of mercury-containing materials, are thermally nonconductive, and bond to tooth structure with the use of adhesives.[4,5] There are problems associated with using resin composite in posterior restorations, however, including shrinkage that occurs on setting[6]; occasional postoperative sensitivity[7,8]; less-than-ideal resistance to wear, particularly if functional cusps are replaced with resin composite; and restoration fracture.[9–11] Minimizing these negative aspects requires meticulous operative technique. Along with appropriate case selection, it is one of the most important variables governing the success of posterior resin composite restorations.[12–14]

Although some questions about longevity remain, there is increasing evidence that properly accomplished posterior resin composite restorations can be quite durable.[10,15,16] Earlier studies comparing the clinical performance of amalgam restorations to resin composite restorations showed amalgam to last longer. Because

studies are often of different duration, it is common to compute an annual failure rate (AFR) to allow a means of comparison. A review from the late 1990s found that high-copper amalgam restorations had an average 1.1% AFR as compared to 2.4% AFR for resin composite.[17] An analysis of more than 300,000 amalgam and resin composite restorations placed in posterior teeth and monitored during a 7-year period in private practices revealed that patients with resin composite restorations had a 16.4% greater chance of restoration failure than those with amalgam restorations at any time period in the analysis.[18] While this appears ominous for resin composite as a posterior restorative material, it should be noted that the probability of a posterior resin composite restoration surviving more than 5 years (93%) differed little from that of an amalgam restoration (94%). More recently, a review of 2,780 Navy and Marine recruits who had satisfactory posterior amalgam or composite restorations upon entry into the military showed that there was a 64% greater risk of composite failure compared with amalgam failure over a period of 3 years.[19] A controlled clinical trial of 1,748 amalgam and composite restorations showed that amalgam had a significantly lower AFR (0.8%) than composite (2.2%) after 7 years.[20] This study also established that larger restorations had a higher failure rate compared with smaller restorations, regardless of restorative material. The risk for secondary caries was 3.5 times greater for composite restorations versus amalgam.

However, a shift in the outcomes of clinical studies comparing amalgam with composite has taken place. Another controlled clinical trial of 1,262 restorations followed for 5 years found no significant difference in the clinical performance of amalgam (3.2% AFR) compared with composite (4.4% AFR). Similar to the previously mentioned clinical trial, larger restorations had a higher failure rate than smaller restorations.[21] Recently, long-term results in practice-based trials have been presented. A 10-year study of Class 1 and Class 2 restorations found no difference in the performance of amalgam (2.1% AFR) versus composite (⅛% AFR) restorations.[16] More recently, a 12-year trial of 1,949 large Class 2 restorations showed that composite performed significantly better (1.7% AFR) than a high-copper amalgam (2.4% AFR). In this study, patients who were considered to have a high caries risk had a significantly greater risk of restoration failure compared with low-caries-risk individuals. Furthermore, in the high-caries-risk patients, three-surface restoration longevity was significantly greater for the amalgam restorations, although there was no difference for four- and five-surface restorations.[22]

When it is used properly, resin composite has demonstrated the ability to perform

as well as amalgam in posterior restorations for up to 10 years.[16,23] Long-term success of resin composite posterior restorations depends on cavity size, restoration type, functional/occlusal stresses on the restoration, patient caries risk status, and tooth type.[20,22,23,24]

There are a number of valuable lessons to be learned regarding the durability of posterior restorations. First, there are a variety of explanations for the different study results based on restorative material and type of study, including patient selection, technique sensitivity, and differences in approach related to practice setting. At the current stage of material development, it is not possible to say that resin composite is a true amalgam replacement capable of providing clinical service to the same level of performance in all of the same clinical situations as amalgam. However, with appropriate case selection and clinical technique, posterior resin composite restorations can serve very acceptably. This chapter presents the factors that will lead to clinical success by examining the advantages, disadvantages, indications, and placement procedures for resin composite as a posterior restorative material.

Advantages of Resin Composite as a Posterior Restorative Material

Esthetics

Manufacturers have developed sophisticated resin composite systems with multiple shades, tints, and opaque resins that allow the practitioner to place highly esthetic restorations (Fig 11-1). Clinical studies often report excellent color matching of resin composite with tooth structure. One study found that 93% of posterior resin composite restorations provided an acceptable color match to adjacent tooth structure after 10 years,[10] while another study found that 94% of resin composite restorations still provided an excellent color match at 17 years.[15] Visible light–cured (VLC) resin composites have less amine content than the autocured systems, resulting in less yellowing of the restoration and greater color stability over time.[25] Microfilled resin composites have the smoothest surface finish of all the systems, and they better maintain their esthetics than other types through enhanced resistance to surface staining.[25]



Fig 11-1 Modern resin composite materials can provide an esthetic restoration, such as the mesio-occlusal restoration in the mandibular first molar. (Courtesy of Bill Dunn, United States Air Force Dental Corps.)

Conservation of tooth structure

In the past, it was recommended that preparation design for posterior resin composite restorations be patterned after the traditional amalgam preparation[14] as described by Dr G. V. Black. Researchers today recommend a more conservative approach.[26,27] The adhesive preparation has evolved to take advantage of resin composite's positive properties and to minimize its negative ones. The current design limits the removal of tooth structure to that needed to eliminate carious tooth structure and fragile enamel.[28]

The adhesive preparation for posterior Class 2 resin composite restorations differs from Black's traditional amalgam design in several ways[29]:

- The preparation tends to be shallower. Because retention is provided through bonding to tooth structure rather than mechanical undercuts, there is no need to penetrate to dentin if the caries lesion does not. This conserves tooth structure and expands the area of enamel available for bonding (Fig 11-2a).

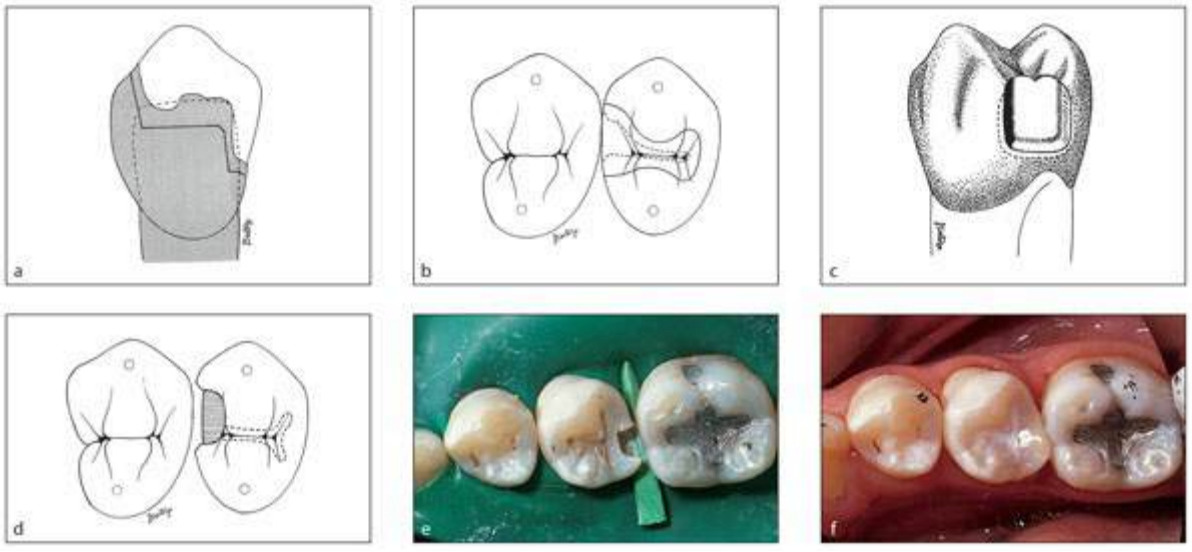


Fig 11-2 *(a)* The adhesive preparation *(upper solid line)* is extended only enough to provide

access and to remove carious tooth structure. It may not penetrate the dentinoenamel junction *(dotted line)*, unlike the traditional preparation *(lower solid line)*. *(b)* The more conservative outline form of the adhesive resin composite restoration *(dotted line)* compared with that of the traditional restoration *(solid line)*. *(c)* In the adhesive preparation, internal angles are rounded. *(d)* If there is no occlusal caries lesion, a Class 2 preparation for a resin composite restoration can be very conservative, similar to a Class 3 preparation. The occlusal fissures can be sealed *(dotted line)* after restoration placement. *(e)* A conservative Class 2 preparation has been made. The occlusal fissures are stained but not carious. *(f)* After the restoration is completed, a sealant is placed in the fissures.

- The preparation tends to have a narrower outline form, which allows less occlusal contact on the restoration and reduces wear.[30,31] A less bulky restoration helps to decrease the adverse effects of resin composite polymerization shrinkage, resulting in improved marginal integrity[28] and less cuspal deflection[32] (Fig 11-2b).
- The preparation has rounded internal line angles; this conserves tooth structure, decreases stress concentration associated with sharp line angles,[25] and enhances resin adaptation during placement[33,34] (Fig 11-2c).
- There is no "extension for prevention" (Fig 11-2d). The occlusal fissures are included in the preparation only if the presence of carious tooth structure dictates this need. Extending the Class 2 preparation through occlusal fissures does not make the restoration more resistant to fracture than the more conservative proximal slot restoration.[26,35] In clinical studies, use of a proximal slot preparation for posterior resin composite restorations showed no failures after more than 2 years of service[36] and 70% success after up to 10 years.[37] In this long-term study,[37] no restorations were lost due to wear or to loss of retention. This may be attributed to the study's requirement that there be an occlusal stop on enamel. Adjacent pits and fissures can be treated with sealants to enhance caries prevention[25] (Figs 11-2e and 11-2f). Figure 11-3 shows some examples of outline form of posterior resin composite restorations; each is designed to treat the pathosis presented, and none has a "standard" shape.

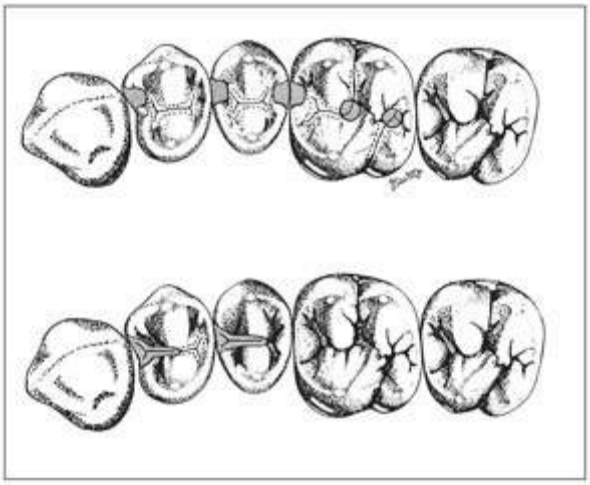


Fig 11-3 Outline of various Class 2 resin composite restorations. Access is limited to that required for removal of carious tooth structure and/or previous restoration(s), so outline form will vary based on the extent of caries lesions. Occlusal pits and fissures that are not carious may be treated with sealants. Deeply stained or demineralized fissures may be opened with a small bur (0.3- or 0.4-mm diameter) or air abrasion prior to sealing. (*Dashed outlines* indicate sealant; *dotted areas* indicate composite.)

Adhesion to tooth structure

The clinical success of bonded resin composite restorations is well documented.[10,15,38,39] The bond between resin composite and tooth structure achieved with bonding systems offers the potential to seal the margins of the restoration[40] and reinforce remaining tooth structure against fracture.[41,42] It has also been suggested that less cuspal flexure occurs with bonded resin composite restorations under subcatastrophic occlusal loads, providing protection against the propagation of cracks that can ultimately lead to fatigue failure.[43] However, not all studies have found that teeth with bonded restorations have an increased resistance to fracture.[43,44] The longevity of the bond is shortened by increased occlusal forces.[45] A study of nearly 11,000 posterior teeth found the prevalence of cusp fracture in posterior resin composite restorations to be no lower than that in amalgam restorations.[46] This supports the fact that bonded posterior restorations cannot be relied upon to provide long-term reinforcement of tooth structure.

Low thermal conductivity

Because resin composites do not readily transmit temperature changes, there is an insulating effect that may help to reduce postoperative temperature sensitivity.[41]

Elimination of galvanic currents

Resin composite does not contain metal and so will not initiate or conduct electrical currents.[25]

Radiopacity

Radiopaque restorative materials are necessary to allow the practitioner to evaluate the contours and marginal adaptation of the restoration as well as to distinguish among the restoration, caries lesions, and sound tooth structure.[47] Studies have shown that the detection of voids and recurrent caries lesions adjacent to restorations is enhanced when the radiopacity of the restorative material is equal to or slightly greater than the radiopacity of enamel.[48–50]

Most, although not all, modern resin composites have a radiopacity in excess of that of enamel.[51,52] For manufacturers to claim that a material is radiopaque, the American Dental Association requires them to demonstrate that a composite has a radiopacity greater than that of an equal thickness of aluminum, which has a radiopacity approximately equal to that of dentin.[53]

Alternative to amalgam

Amalgam, despite having a long track record of clinical success,[54–56] has declined in use as a restorative material primarily because of its unesthetic appearance but also because of its mercury content. Concerns about mercury in amalgam are more psychologic than scientific, as evidenced by a comprehensive literature review of 950 scientific and medical studies that found no correlation between dental amalgam and health problems.[57] Likewise, two extensive controlled clinical trials found no evidence of adverse health effects in children who received amalgam restorations compared with those who received composite restorations over 5 years[21] and 7 years.[20] Despite this, there is an increasing desire to find mercury-free alternatives to dental amalgam.[58] Patients are aware of indictments against amalgam, and some express concern about potential health hazards.[59] Amalgam is also less attractive to dental professionals as government agencies consider classifying it as hazardous waste[60] and requiring that dental offices install expensive separator systems to remove mercury from wastewater.[58,61] As a result, resin composite use in posterior restorations continues to gain popularity in the profession.[58]

Disadvantages of Resin Composite as a Posterior Restorative Material

Polymerization shrinkage

Despite improvements in resin composite formulations over the years, modern systems are still based on variations of the bisphenol glycidyl methacrylate (bis-GMA) molecule, which has been used for more than 40 years.[62] One of the major drawbacks of this material is the polymerization shrinkage that occurs during the setting reaction. Modern resin composites undergo volumetric polymerization shrinkage of 1.5% to 5%.[63–65]

Most of the problems associated with posterior resin composite restorations can be related directly or indirectly to polymerization shrinkage and the stress that develops during polymerization. During polymerization, resin composite may pull away from the least retentive cavity margins (usually those with little or no enamel on them), resulting in gap formation.[66–68] Tensile forces developed in enamel margins can result in marginal degradation from mastication.[69] Contraction forces on cusps can result in cuspal deformation,[70] enamel cracks and craze lines[71] (Fig 11-4), and, ultimately, decreased fracture resistance of the cusps.[72]

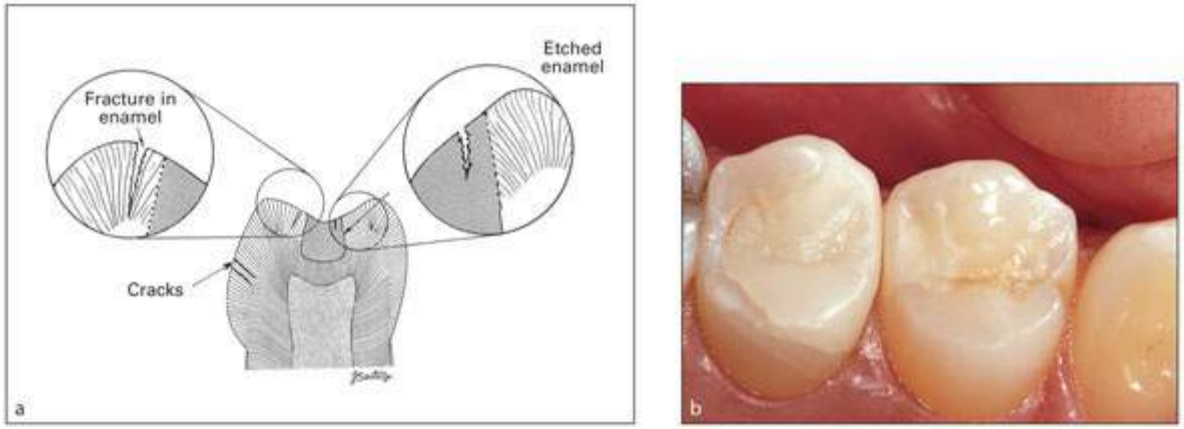


Fig 11-4*(a)* Polymerization shrinkage can cause crazing in the enamel or fractures within the resin composite. *(b)* Craze lines are evident in the lingual cusp of the maxillary right second premolar bonded to a very large Class 2 resin composite restoration.

Polymerization shrinkage occurs regardless of the system used to initiate the setting reaction. For many years, it was believed that autocured resin composite polymerizes toward the center of the mass of the resin composite, while VLC resin composite polymerizes toward the light source.[6,68] Research has provided evidence

that polymerization shrinkage occurs toward the walls of cavity preparations to which it is bonded most strongly, regardless of the initiator mode.[73,74]

A number of techniques have been suggested to decrease the adverse effects of polymerization shrinkage. The most commonly used technique is incremental placement of VLC resin composite, which decreases the effect of setting contraction by reducing the bulk of resin composite cured at a time.[68] In addition, incremental insertion reduces the ratio of bonded to unbonded surface area, which helps to relieve the stress developed at the bond between tooth and resin composite.[75] The incremental placement technique is discussed later in this chapter.

Autocured resin composites are sometimes recommended for posterior restorations, because an autocured composite tends to induce less polymerization stress than does a comparable bulk of VLC composite. This is due in part to greater porosity being incorporated into the autocured resin composite as a result of mixing. The incorporated oxygen inhibits the set of resin immediately adjacent to the voids and decreases the ratio of bonded to unbonded surface area.[76] The voids increase the free surface area for stress compensation by flow of the resin during the setting reaction.[77] In addition, because of a slower polymerization rate, autocured resin composites develop shrinkage stresses more slowly than do VLC materials. This allows for increased restorative material flow during polymerization.[78,79] However, a number of problems associated with the use of autocured resin composite in posterior restorations argue against its use. These problems are discussed in subsequent sections.

Decreasing the rate of polymerization of VLC resin composite can be accomplished by varying the intensity of the curing light. This has resulted in a form of curing for VLC resin composites variously referred to as *two-step* or *soft-start polymerization*. Some studies have shown that by reducing the initial irradiance to approximately 150 mW/cm^2, followed by high-level irradiance (650 mW/cm^2 or greater), the curing reaction is slowed, marginal integrity is enhanced, and mechanical properties are not adversely affected.[80–84] However, results with these reduced-irradiance curing regimens have been equivocal,[85] and it is difficult to assess whether or not the laboratory results will have any relation to clinical outcomes. Research has demonstrated that a truly significant decrease in polymerization shrinkage stress may require what many consider to be clinically impractical curing regimens.[85]

The best hope for overcoming the problems of polymerization shrinkage lies in the future development of tooth-colored materials that contract minimally during setting,

an area of vigorous research.[86–88] While it is intuitive to think that producing a material with zero shrinkage is ideal, this may not be true, because dental composites are based on polymers that absorb small amounts of water, the result of which is a slight volumetric expansion that may apply pressure to the cavity walls.

Secondary caries lesions

Several clinical studies and reviews of clinical studies have demonstrated that secondary caries is a significant cause of failure of posterior resin composite restorations.[20,89,90] It is believed that the marginal gap formed at the gingival margin as a result of polymerization shrinkage allows the ingress of cariogenic bacteria (Fig 11-5). Because marginal degradation has been demonstrated to increase with time,[91,92] the risk of secondary caries also increases with time.



Fig 11-5 *(a)* In these posterior resin composite restorations, the dark shadowing adjacent to the occlusal margins was caused by recurrent caries lesions. *(b)* After placement of a rubber dam, the resin composite restorations are removed and caries-detecting solution is placed. *(c)* Stained areas confirm the presence of demineralized tooth structure.

Studies have shown that levels of *Streptococcus mutans*, the organism linked most closely to the production of dental caries lesions,[93] are significantly higher in the plaque adjacent to proximal surfaces of posterior resin composite restorations than in plaque adjacent to either amalgam or glass-ionomer restorations. In addition, the organic acids of plaque have been found to soften bis-GMA polymers, and this in turn could have an adverse effect on wear and surface staining.[94] These facts emphasize the need for regular recall and close follow-up of patients with posterior resin composite restorations.

Postoperative sensitivity

Postoperative sensitivity has been associated with the placement of posterior resin composite restorations. Earlier clinical research noted postoperative sensitivity rates as high as 29% following placement of the restorations.[7] Some studies have continued to report postoperative sensitivity following the placement of

even small posterior resin composite restorations,[36,95] but reports of postoperative sensitivity have diminished with improvements in dentin adhesives and as dentists have become more proficient in the case selection and clinical techniques associated with successful posterior composite restorations.[16,89,95–97]

A number of reasons have been postulated for the occurrence of postoperative sensitivity, but the most commonly accepted theories relate to polymerization shrinkage. As previously discussed, polymerization shrinkage results in formation of a gap, which allows bacterial penetration and fluid flow within it. The bacteria or their noxious by-products may enter the dentinal tubules and cause pulpal inflammation and tooth sensitivity.[98] Gap formation also allows a slow, continuous outflow of dentinal fluid from the pulp through the tubules to the gap. Cold or other stimuli may cause a contraction of fluid in the gap, leading to a sudden, rapid outflow of tubular fluid that stimulates the nerve surrounding the odontoblastic processes and results in the perception of pain.[99]

Contraction forces of polymerization shrinkage may also result in cuspal deformation, with resultant cracking and crazing of remaining tooth structure (see Fig 11-4b), which can cause tooth sensitivity.[66] Flexure of resin composite under an occlusal load may cause hydraulic pressure in the tubular fluid to be transmitted to the odontoblastic processes, another possible cause of tooth sensitivity.[66,99]

Postoperative sensitivity can be a disturbing sequela for both patient and dentist. Obviously the discomfort associated with restoration placement is unpleasant for the patient. Likewise, it is often frustrating for the dentist to try to determine the cause and eliminate postoperative sensitivity. This frustration is increased with the awareness that patients who experience postoperative sensitivity within 1 month of posterior resin composite restoration placement are about twice as likely to have that restoration fail within 5 years as those who do not experience postoperative sensitivity. This risk increases as the size of the restoration increases.[8] Awareness of the potential for postoperative sensitivity allows the practitioner to forewarn the patient of this possibility. Careful adherence to the guidelines for case selection and restoration placement, including the rebonding procedure described later in this chapter, will help to reduce this problem.

Decreased wear resistance

The wear resistance of posterior resin composite restorations has been the subject of considerable attention. This characteristic has long been a concern for restoration

longevity,[100] although it has improved as refinements in materials have occurred.[16,96] In reports of clinical studies, it has been noted that no posterior resin composite restorations were replaced as a result of excessive wear after 7 years[101] and only one was replaced as a result of excessive wear in a 10-year study.[23] A practice-based retrospective study showed no failures in 1,955 posterior composite restorations due to wear after 10 years.[16]

Resin composite wear results from the combination of chemical damage to the surface of the material and mechanical breakdown.[102–105] Resin composites undergo wear by two different mechanisms: abrasion and attrition. *Abrasion* is generalized wear that occurs across the entire occlusal surface of the resin composite as a result of the abrasive action of particles during mastication (Fig 11-6). This type of wear occurs in all areas of the restoration. *Attrition* is the loss of material that occurs as a result of direct contact with opposing tooth surfaces in the occlusal contact areas of the restoration[106] (Fig 11-7). Generally, wear can be related to either material or clinical factors.

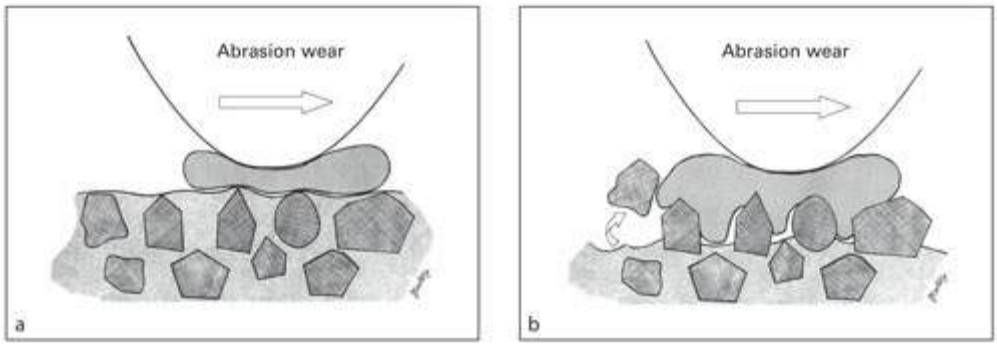


Fig 11-6 Abrasion wear occurs across the entire occlusal surface of the resin composite. *(a)* The softer resin is preferentially worn away, exposing the harder filler particles. *(b)* Eventually, enough of the filler particle is exposed so that it is "plucked" from the surface of the resin composite.

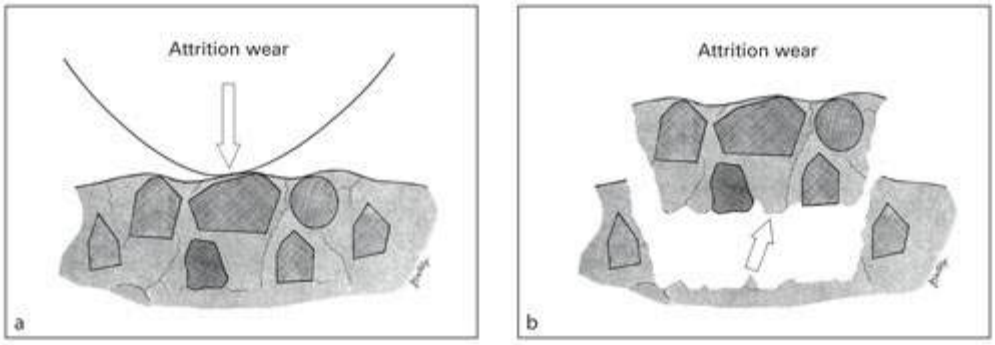


Fig 11-7 Attrition wear occurs in occlusal contact areas. *(a)* Cracks occur in the resin matrix as a result of occlusal stress. *(b)* Eventually, the cracks coalesce and result in loss of resin composite material from the surface.

Material factors relate primarily to the content, size, and distribution of the resin composite's filler particles.[107] Clinical studies, in general, have shown less heavily filled composites (less than 60% filled by volume) to exhibit unacceptable wear.[100] Because of their generally lower filler content (30% to 50%), microfilled composites are more subject to attrition[69] and marginal breakdown,[91,92,108] especially adjacent to occlusal contact areas.[69] However, they are more resistant to abrasion because of their smoother surfaces, decreased interparticle spacing, and lowered coefficient of friction.[109,110] The more heavily filled hybrid resin composites are more resistant to attrition than are the microfilled materials.[108] However, resin composites that have a larger mean particle size (greater than 3 μm) tend to have significantly higher abrasion wear.[31,88,100] This is due to the preferential wear of the resin between the fillers, which causes the filler to become loosened and ultimately lost from the resin matrix, leading to three-body wear and generalized loss of material[100] (Fig 11-8). A newer nanofilled composite formulation is providing in vitro and in vivo wear characteristics similar to microhybrid composites.[110,111]



Fig 11-8 A Class 2 lesion previously restored with a larger-particle resin composite. Pitting and generalized wear of the occlusal surface can be seen as a result of abrasion wear. In addition, interproximal abrasion wear has resulted in the loss of interproximal contact on the mesial.

The rate of wear varies with particle size as well. As mean filler particle size decreases (below 1 μm), the wear rate tends to be linear with time. Conversely, composites with larger filler particles tend to have more rapid wear initially but slower wear over time.[88,91,92,112,113]

Clinically relevant wear factors include the size of the restoration, its location in the arch, the occlusal load it must withstand, and how well the resin composite is cured. As the surface area and length of cavosurface margins increase, so does the exposure to occlusal forces, with a resultant increase in wear.[16,20,21,30,31,97,111] The more posterior a tooth, the greater the masticatory forces and the more rapid the wear of the restoration.[30,96,97,111,114] Fracture resistance decreases as a result of

fatigue from chewing,[115] and increased chewing pressure will result in increased wear.[116] Finally, if the clinician does not properly cure the composite material, the degree of conversion of the composite will suffer, and wear will increase.[108]

Proximal surfaces are subjected to the forces of abrasion during function as well, due to individual tooth movement during mastication that causes wear in the interproximal contact areas. Early research showed that proximal surface wear of posterior resin composite restorations was significantly greater than that of enamel proximal surfaces.[117,118] One study even found proximal wear rates to be higher than occlusal wear rates in resin composite restorations,[119] possibly due to increased contact time between contacting surfaces, differences in cure, or different environments. However, later studies do not indicate that proximal surface wear due to interproximal contact is a problem with current resin composites.[101,120,121]

Clinical studies show that resin composite formulations have acceptable wear characteristics for up to 17 years.[15,23,96,97] While some research reports that resin composites have significantly higher wear rates than amalgam,[106,122] other studies have indicated that posterior resin composite restorations resist wear as well as amalgam restorations,[23,104] including recent large practice-based trials.[16,22]

Other mechanical properties

Generally, the more closely the mechanical properties of a restorative material simulate those of enamel and dentin, the better the restoration's longevity.[88,123,124] A number of the mechanical properties of resin composite are inferior to those of tooth structure and of other restorative materials. These inferior properties can have an adverse effect on the durability of the restoration.

Resin composite materials have low fracture toughness in comparison with metallic restorative materials.[67] Indeed, bulk fracture of posterior resin composite restorations has been noted as a significant cause of failure in many clinical studies.[20,21,90,96,97] Increased filler loading of resin composite leads to improved fracture toughness.[103,115] Resin composite has a relatively high degree of elastic deformation (ie, a low modulus of elasticity) that is six to eight times that of amalgam.[99] Failures of resin composite restorations associated with the high elastic deformation of the material have included bulk fracture,[67] microcrack formation,[103] and relatively low resistance to occlusal loading.[7] As with fracture resistance, more highly filled composites exhibit less elastic deformation than their less-filled counterparts.[88]

The coefficient of thermal expansion of resin composite is another property that differs significantly from that of tooth structure.[125,126] Because the coefficient of thermal expansion of resin composite is higher than that of tooth structure, composite tends to expand and contract more than enamel and dentin when subjected to variations in temperature. This can increase marginal gap formation and exacerbate the effects of polymerization shrinkage on cuspal deformation, and it may result in the fracture of composite or enamel at the margins.[67,124] It has been demonstrated that as the mismatch in thermal-expansion properties between restorative material and the tooth structure increases, so does marginal leakage.[127] As the filler content of resin composite increases, however, the mismatch decreases.[125]

Water sorption

Water sorption is another factor influencing the clinical performance of resin composites. Water is absorbed preferentially into the resin component of resin composite materials, and water content is therefore increased when resin content is increased.[62,67] Because of the swelling of the resin matrix from water sorption, the filler particle bond to resin is weakened. If the stress is greater than the bond strength, the resulting debond is referred to as *hydrolytic breakdown*.[67,128] Incompletely cured resin composite will exhibit more water sorption and greater hydrolytic degradation.[129]

It has been suggested that the swelling of resin composite caused by water sorption can be beneficial due to closing of the marginal gap caused by polymerization shrinkage. However, studies have shown that the swelling from moisture absorption usually is not enough to overcome the polymerization shrinkage gap.[69] Even if water sorption did result in a closed marginal gap, it would only provide a close adaptation, without adhesion between the resin composite and tooth structure.[130]

Variable degree of conversion

Analysis of the amount of polymerization and cross-linking, or *degree of conversion*, of resin composites reveals that certain characteristics of this material are at odds with one another. As the degree of conversion of a resin composite material increases, the mechanical properties improve.[62,76,109] Clinical research has clearly demonstrated that a reduced degree of conversion causes significantly increased wear.[108] However, polymerization shrinkage also increases with more

conversion.[131] Resins with decreased filler content exhibit lower viscosity and improved diffusion of reactive groups during the polymerization reaction and, thereby, improved cure.[132] However, a decreased filler content also results in inferior mechanical properties[88,103,115,125] and poorer clinical performance.[7,69,91,92] Achieving the best balance among these factors is a challenge for both manufacturers and clinicians.

VLC composites have been shown to achieve a somewhat higher degree of conversion than autocured materials.[132,133] Several factors influence the extent of polymerization of VLC composites. Lighter shades cure more easily and with less illumination time than darker shades.[6,134] Resin composites with larger (greater than 1.0 μm) or very small (less than 0.1 μm) filler particles tend to transmit light throughout the material more effectively than those with filler particles that are close in size to the wavelength of the curing light (~0.5 μm).[134,135] The longer the composite is subjected to the curing light, the more effective the cure,[136] but the thickness of each increment should be limited to 2.0 mm.[6,134] The degree of conversion is inversely related to the distance of the light tip from the resin composite,[136] and tip distances greater than 6.0 mm from the surface of the increment can significantly decrease resin composite cure.[137] Of course, the condition of the curing unit can, also impact the effectiveness of the cure.

Inconsistent dentin adhesion (marginal leakage)

Polymerization shrinkage causes the resin composite to pull away from cavity margins, resulting in gap formation.[66–68] Despite advances in dentin adhesive systems, they still do not consistently and reliably achieve bond strengths to dentin and cementum that are high enough to prevent this occurrence[138–141] (Fig 11-9; see also Fig 13-13). This sometimes results in open margins, sensitivity, interfacial staining, and bacterial invasion.[142] In addition, the bond between adhesive and tooth structure has been shown to degrade with aging, both in vitro and in vivo.[141,143–146]



Fig 11-9 Marginal staining/discoloration indicating degradation of adhesion between the resin composite and the cavity preparation that occurs with aging. Facial Class 5 resin composite restorations in the maxillary right first and second premolars after 3 years. Discoloration is evident, particularly at the gingival (dentin) margins of the restorations.

Technique sensitivity

Because of the negative aspects of using resin composite as a posterior restorative material described previously, one of the most important variables in clinical success is the placement technique.[13] Application technique has been shown to significantly affect adhesive bond strength.[147] Technique may also account for the great variability reported in clinical success rates for posterior resin composite restorations. The meticulous operative procedures demanded for placing these restorations require increased chair time. Clinical research has shown that posterior resin composite restorations require significantly more time to place than do comparable amalgam restorations.[23]

Indications for Resin Composite as a Posterior Restorative Material

Resin composite is now the state-of-the-art restorative material for direct-placement posterior restorations. It is popular with patients because of its esthetic and metal-free (especially mercury-free) characteristics. The esthetic and mechanical properties of modern resin composites have considerably improved since their introduction. Dentists use more resin composite than amalgam for direct-placement restorations,[148] and this discrepancy between the use of amalgam and resin composite will likely only increase in the future. Results of clinical studies demonstrate that resin composite can fare comparably with amalgam when used in posterior restorations.[16,21,22] Based on these facts and the previous discussion, resin composite is the first choice for direct-placement posterior restorations. The only absolute contraindication is if the patient is allergic or sensitive to resin-based materials. Otherwise, the clinician should consider other options only as clinical conditions dictate, including the following relative contraindications for resin composite use in posterior teeth:

- The restoration will replace supporting cusps. Extensive resin composite restorations including the supporting cusps of posterior teeth do not perform as

well as other restorative materials[11] (Fig 11-10).



Fig 11-10 Defective mesio-occlusodistal (MOD) amalgam restoration in the mandibular second premolar. Excessive width and need for cusp replacement precludes this amalgam restoration from being replaced with resin composite. A discolored fracture line and recurrent caries running from the MOD amalgam to the amalgam in the facial cusp tip would necessitate functional cusp coverage with the replacement restoration.

- Large restorations are required in the teeth of a patient with significant parafunction, bruxism, or heavy occlusal stress, where most or all function will be on the restoration and not on tooth structure.[9,12,15,24,37]
- The patient has a high caries risk and is noncompliant with oral hygiene instructions. Secondary caries is a significant cause of posterior composite failure.[20,22,39,149]
- The clinician is unable to obtain adequate field isolation.[150,151]

Also, in regard to resin composite restorations with margins in thin enamel near the cementoenamel junction or on cementum or dentin, the use of the *open sandwich technique* (also referred to as the *bonded-base technique* in Class 2 restorations) has been shown to provide acceptable clinical performance,[152–155] providing significant protection of the tooth structure near the gingival margin against demineralization.[152] In the open sandwich technique, a restorative resin-modified glass ionomer (RMGI) is used in the portion of the restoration near the gingival margin. In a Class 2 restoration, that would be the gingival increment of material placed in the proximal box. If this technique is to be used, however, it is important that the patient be at low risk of developing caries lesions. The main benefit of the open sandwich technique is an improved marginal seal and protection of the gingival margin against demineralization, but this benefit can be overwhelmed by increased plaque formation. A clinical trial evaluating the use of this technique found that 67% of recurrent caries lesions adjacent to the RMGI portion of an open sandwich

restoration occurred in subjects with poor oral hygiene.[154] For a description of the bonded-base technique, see the section on Class 2 resin composite restorations in this chapter.

Informed Consent

It is incumbent upon the dentist to provide the patient with appropriate information regarding the restorative procedures being recommended. This should include a description of the proposed procedures in lay language, any risks associated with the procedures, and other options the patient may wish to consider. In addition, the patient should be provided the opportunity to have his or her questions answered. This is sometimes referred to as the PARQ process (Procedures, Alternatives, Risks, Questions). Dlugokinski and Browning[156] surveyed dentists and patients regarding their respective attitudes and preferences toward informed consent. The survey focused on offering patients the alternatives of resin composite versus amalgam for direct-placement posterior restorations. The survey revealed that as a dentist performed more resin composite restorations compared with the number of amalgam restorations performed, the dentist was less likely to discuss the adverse aspects of resin composite.[156]

Conversely, the survey revealed some intriguing insights from the patient's perspective[156]:

- 100% of patients wanted the dentist to tell them about all aspects of alternative restorative materials.
- 75% of patients preferred a written explanation rather than simply an oral explanation.
- 82% of patients would not choose a posterior resin composite restoration to replace an adequate amalgam restoration.
- Patients were equivocal over cost as a factor for selecting amalgam versus resin composite: 43% would still choose a resin composite restoration even if it cost 50% more than an amalgam restoration and would not last as long (versus 36% preferring amalgam).

The results clearly show that patients do not want the dentist alone to decide which restoration will be placed. While most patients desire the dentist's recommendation, the ultimate decision is theirs, and they want to be fully informed

to make the best choice for their particular situation. Most patients want information in writing. Several commercial companies and the American Dental Association (ADA) offer pamphlets that explain the indications, advantages, and disadvantages of various restorative materials. The ADA also provides a document with information on various restorative materials that can be downloaded at no charge.[157]

Autocured versus Light-Cured Resin Composites

Autocured resin composite restorative materials largely disappeared from clinical practice in the 1980s because of the popularity of the light-cured materials. Some clinicians have recommended the use of autocured resin composites, either alone or in combination with VLC resin composites, for posterior applications.[158–160] The primary advantage of an autocured material is that it can be placed in bulk, saving time compared with the incremental insertion technique used with VLC materials.

Although more time-consuming, use of VLC resin composites has a number of advantages over the use of autocured resin composites. VLC composites achieve more complete polymerization,[133] resulting in superior mechanical properties,[62,71,137] and they exhibit better color stability.[161] Autocured composites tend to incorporate voids as a result of mixing two-paste systems,[77] and the increased porosity decreases tensile strength and surface smoothness,[162] accelerating wear.[163] Extended mixing and manipulation may interrupt the polymerization process and may compromise the integrity of the polymer network, resulting in reduced strength and wear resistance.[164]

VLC composites should be used with an incremental placement technique to reduce polymerization shrinkage stress and achieve optimal polymerization in the definitive restoration.[165] This technique allows the practitioner to build up and sculpt the restoration. Research has shown VLC composite increments to have adequate interfacial strength.[166] Perhaps most important, VLC resin composites performed better in clinical trials than autocured materials over 1 year[167] and 3 years.[117]

Direct Posterior Resin Composite Restorations

Preoperative evaluation

The factors noted as contraindications in the previous list should be considered in the preoperative evaluation. The occlusion should be marked with articulating paper as a guide to preparation design. The best type of resin composite for the restoration should be chosen. At present, the heavily filled small particle–hybrid composites (sometimes known as *micro-hybrids*) are considered best suited for posterior use.[12,16,22,88,90]

For many posterior resin composite restorations, shade selection is not critical. In fact, some clinicians prefer a deliberate shade mismatch to aid in subsequent finishing and future evaluation procedures. But when shade is important, shade selection should be performed before isolation of the tooth, because isolated teeth become dehydrated, and dehydration changes the shade of the enamel. A shade is chosen from the shade guide that accompanies the composite, and then a small portion of the composite is placed on the unprepared and unetched tooth and polymerized (Fig 11-11). The resin "test shade" can easily be removed because the tooth surface has not been etched or primed prior to its placement. See chapter 4 for more information on shade selection.



Fig 11-11 A small amount of resin composite is placed on the unprepared tooth to verify the shade prior to isolation with a rubber dam.

If the dentist is going to use a warm composite placement technique, an appropriate amount of resin composite may be transferred to a syringe tip (Centrix) that is amber-colored or opaque to prevent premature polymerization. The Centrix syringe tip is then placed in a composite warming tray (Calset, AdDent) at 60°C to 68°C (140°F to 155°F) (Fig 11-12). This will reduce the resin composite's viscosity and aid in subsequent placement. The rationale for this technique is discussed in the section on Class 2 resin composite restorations later in this chapter.



Fig 11-12 An appropriate resin is selected and placed into a light-protected syringe tip, and the protected syringe tip is put into a warmer (60°C to 68°C; 140°F to 155°F) to reduce viscosity.

Isolation

Securing and maintaining control of the operative field is essential to the success of the restoration. One clinical study has demonstrated no difference in the performance of posterior resin composite restorations whether or not a rubber dam was used during restoration placement.[168] A large retrospective practice-based study showed a high success of posterior composites even though a rubber dam was rarely used.[16] However, in another clinical study, the margins of all Class 2 resin composite restorations placed without a rubber dam demonstrated marginal leakage 4 to 6 weeks after placement.[169] With the advent of adhesive systems that use more hydrophilic components, research has shown that contamination with moisture or saliva affects some adhesives but not others.[170–172] Blood contamination will adversely affect adhesion in all bonding systems.[171–173] In the authors' opinion, the most reliable method to accomplish field isolation is the placement of a rubber dam. The rubber dam prevents moisture contamination and protects gingival tissues from laceration[169] (Fig 11-13).



Fig 11-13 The occlusion is marked and rubber dam isolation achieved. The rubber dam has been inverted around the teeth to ensure moisture control.

Sealants and Preventive Resin Restorations

Resin sealants

While not normally considered to be posterior resin composite restorations, resin fissure sealants have been in use as a preventive restorative procedure for several decades. Sealants provide an effective means of reducing the incidence of caries lesions in fissures.[174] Compared to teeth with unsealed fissures, teeth with resin-sealed fissures have demonstrated a 35% reduction in fissure caries lesions during a

5-year period,[175] a 43% reduction over 4 years,[176] and a 55% reduction over 7 years[177]. However, there are a number of factors that must be considered regarding fissure sealant effectiveness. Numerous clinical studies have demonstrated that sealants tend to fail at a rate of 5% to 10% per year.[178–181] This is significant because the caries rate for teeth in which sealants are partially or totally lost increases significantly, in many cases equaling the caries rate of unsealed teeth.[175–177,179,180] The key to resin sealant success in preventing caries lesions is total retention of the sealant.[174,175,177,180] Enhancing complete resin sealant retention will therefore enhance the caries-reduction benefit. Some factors that affect sealant retention and effectiveness include the following:

- Mandibular teeth show higher retention rates than maxillary teeth; premolars show higher retention rates than molars.[179,182]
- Annual recall of patients and repair of partially or totally lost sealants improves effectiveness.[179,181]
- Use of bonding agents following phosphoric acid etching and prior to sealant placement helps to wet fissures,[183] improve sealant penetration into fissures,[184] increase bond strength,[185,186] improve sealant adhesion to saliva-contaminated enamel,[185–187] and improve clinical retention of sealants.[174,188–190]
- Individual studies have demonstrated that light mechanical preparation of fissures with a very small bur (0.3- to 0.4-mm diameter, rounded tip) or air abrasion can provide clinical advantages, including exposing sound, unstained enamel prior to etchant placement; enhanced sealant penetration and attachment; decreased bubble formation; improved marginal adaptation[191,192]; decreased marginal leakage[193]; improved microbial elimination[194]; and increased clinical retention compared with unprepared fissures.[192] However, two systematic reviews concluded that the evidence for the benefit of mechanical preparation was conflicting and limited.[174,195]
- Clinical studies of RMGI sealants show good caries prevention but very poor mechanical retention compared with resin sealants. However, the caries-prevention benefits of RMGI sealants are comparable with those of resin sealants and should be considered when moisture control may compromise resin sealant retention.[174]
- Flowable resin composite materials have been shown to perform as well as fissure sealants. One short-term clinical study demonstrated comparable

performance for a flowable resin composite and a sealant,[196] while another showed improved performance of a flowable resin composite material when compared with a traditional resin sealant.[197]

- The level of caries activity is critical to the cost-effectiveness of sealants; if a patient exhibits a low caries index, then the value of this procedure is low.[198] Clinical parameters including occlusal fissure morphology; number of decayed, missing, or filled surfaces (DMFS); and the clinician's subjective judgment are all significant predictors of future occlusal caries activity.[199,200] Therefore, sealant use should be based on a determination of the patient's disease level and his or her potential for future fissure caries lesions; sealants should not be placed universally.[174]

Preventive resin restorations

A restoration that maximizes the benefits of conservative adhesive dentistry is the *preventive resin restoration* (PRR). Suggested by Ulvestad[201] in 1975 and popularized by Simonsen[202,203] and Simonsen and Stallard,[204] the PRR was developed to overcome problems associated with traditional "extension for prevention" in restorations necessitated by minimal occlusal caries lesions.

The PRR limits preparation to pits and fissures that are carious. Once the lesion is eliminated, no further preparation is performed. If the resultant preparation is restricted to a narrow and shallow opening of the fissure, a resin sealant (or flowable resin composite material) is placed. If additional tooth structure is removed, a posterior resin composite is placed in that area, and the remaining fissures and the surface of the resin composite restoration(s) are sealed with resin sealant material or flowable composite. A number of advantages have been ascribed to this technique, including the following:

- Conservation of tooth structure: One 5-year clinical study determined that the average occlusal amalgam occupied 25% of the occlusal surface compared to just 5% for an average PRR.[205]
- Enhanced esthetics: This is provided by the tooth-colored restorative material.[206]
- Improved seal of restorative material to tooth structure: This is achieved through the bond of resin composite to etched enamel with an adhesive resin.[207]
- Minimal wear: This is due to restricted cavity-preparation size, so that occlusal contacts on the restoration are limited.[208]

- No progression of sealed caries lesions: If a caries lesion is inadvertently allowed to remain in or at the base of a sealed fissure, it will not progress, because the sealant prevents nutrients from supplying cariogenic bacteria.[209]
- Good longevity: Clinical studies have demonstrated that PRRs are successful for periods of up to 10 years[206–210] and can equal[205] or exceed the performance of amalgam restorations.[209]

The same provisos concerning sealants must be applied to PRRs; that is, the sealants placed in association with PRRs will tend to be lost at a rate of 5% to 10% per year.[206,208,210] Therefore, these restorations must be monitored over time, and the sealants and/or restorations must be repaired or replaced as needed.[206,211]

Indications and contraindications

A PRR is indicated when some areas of the fissure system of a tooth are associated with carious dentin and others are not. The extent of the anticipated restoration should be such that occlusal forces will be limited primarily to tooth structure.

Technique

The preoperative evaluation, marking of occlusion, shade selection, and rubber dam isolation should be accomplished first (Fig 11-14a). After the resin composite is selected, it may be used directly from a unit-dose ampule; if it is in a syringe, it should be placed into a light-protected syringe tip (see Fig 11-12). If desired, the viscosity of the resin composite may be further reduced by heating it in a composite warming tray to enhance adaptation to the cavity preparation as described in the section on Class 2 restorations.

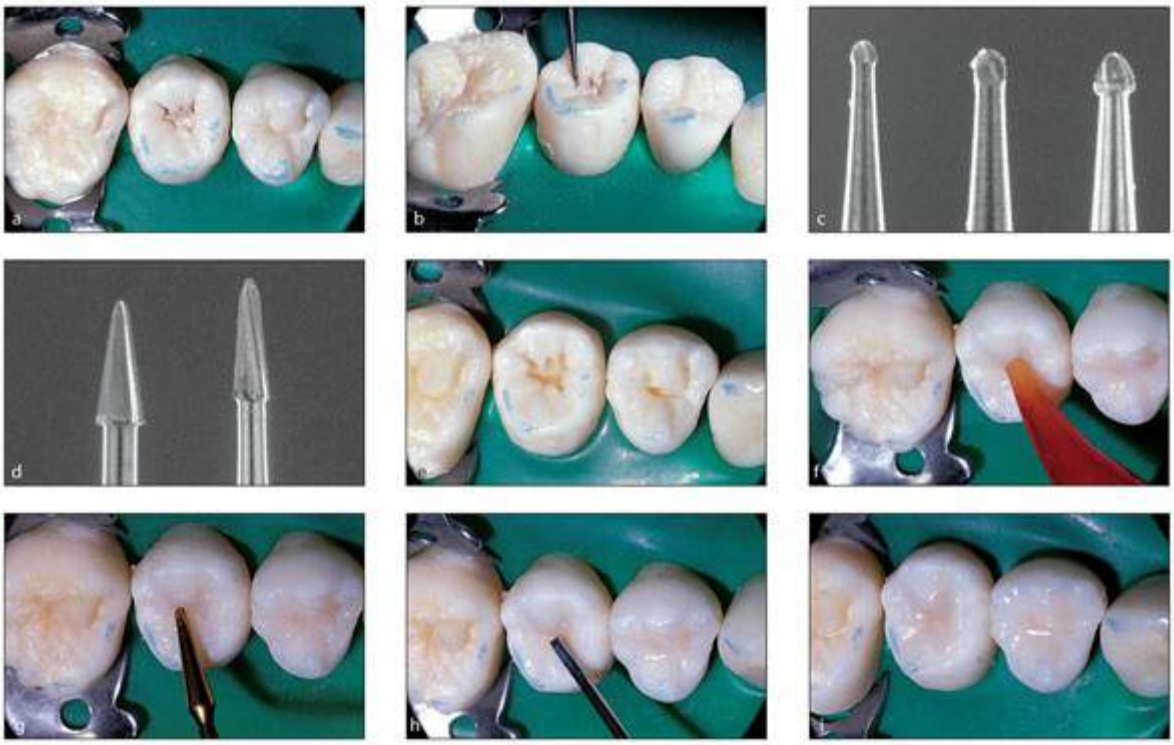


Fig 11-14 Technique for placing preventive resin restorations. *(a)* The occlusal surfaces of the maxillary first and second premolars exhibit demineralization of deep occlusal fissures. A rubber dam has been placed and inverted. The occlusion was marked prior to rubber dam placement. *(b)* Preparation is initiated with a small (no. 1/8) round bur in a high-speed handpiece. *(c)* 1/16, 1/8, and ¼ round burs (0.3, 0.4, and 0.5 mm in diameter, respectively) work well to open occlusal fissures. *(d)* Alternatively, fissurotomy burs (SS White) can be used to open occlusal fissures. *(e)* Completed preparations, which were limited to access for removal of carious tooth structure. The preparation of the first premolar was limited and mostly confined to fissure enameloplasty. Penetration to dentin was minimal in the second premolar, occurring in only limited locations. *(f)* After adhesive system placement, restorative resin composite is syringed into the deeper (greater than 0.5 mm) areas of the preparation. *(g)* A hand instrument, such as the PKT3 shown, is used to develop occlusal anatomy in the final resin composite increment prior to curing. *(h)* The occlusal surface is re-etched, and occlusal sealant is placed over the resin composite and any remaining pits and fissures. *(i)* Completed occlusal preventive resin restorations.

The conservative adhesive preparation eliminates demineralized dentin, overlying unsupported enamel, and associated demineralized enamel. The preparation should be initiated with the smallest instrument that will accomplish this limited preparation, such as a no. 1/16, ⅛, or ¼ round bur (Figs 11-14b and 11-14c), fissurotomy bur (Fig 11-14d), or air abrasion. Larger instrumentation is used only as the size of the caries lesion dictates. Any fragile or unsupported enamel remaining on the occlusal surface after removal of the demineralized dentin should be removed. No bevels should be placed on the occlusal margins of the preparation.

Typically, these restorations are limited in size and depth, so no pulpal protection is needed (Fig 11-14e), but a PRR is indicated even with a deep caries lesion when

areas of the fissure system are not carious. Etching and application of an adhesive agent are the same as with other adhesive restorative procedures. Those areas of the preparation that have extended into dentin are filled with a highly filled restorative resin composite, which should be cured in increments no greater than 2 mm in thickness (Fig 11-14f). If the preparation is narrow and shallow, sealant material, flowable resin composite, or warmed resin composite may be used. Prior to curing of the final increment, occlusal anatomy is sculpted with a hand instrument (Fig 11-14g). Occlusal sealant is then placed over the resin composite, through remaining prepared or unprepared etched fissures, and cured (Fig 11-14h). If, after the final increment is cured, excess restorative material is present, the surface should be adjusted to provide desired contours and anatomy. The entire fissure system, including the resin composite and sealant, should be re-etched and a surface-sealing resin applied.[31,212]

After completion of the restoration (Fig 11-14i), the rubber dam is removed and correct occlusion verified or obtained.

Other Class 1 Resin Composite Restorations

When a Class 1 restoration is being placed because of initial caries lesion(s), the PRR is usually the technique of choice. If there was a previous restoration, the outline form and depth of the preparation will be determined by the previous restoration and any new pathosis. Margins of occlusal preparations for resin composite should not be beveled.[213] Lining and bonding techniques should be used as described for Class 2 restorations.

Class 2 Resin Composite Restorations

As with PRRs, Class 2 restorations should be limited to obtaining access to the carious dentin and removing it and any overlying fragile or demineralized enamel.

Prewedging

Obtaining adequate interproximal contact in the definitive restoration starts at this stage of the restorative procedure, not at matrix placement or material insertion, as is the norm for amalgam restorations. Uncured resin composites, even the so-called packable composites, do not have the ability to hold the matrix band in close

adaptation to an adjacent tooth.[214,215] This makes obtaining an adequate interproximal contact one of the more difficult aspects of placing a Class 2 resin composite restoration. Placement of an interproximal wedge at the start of the procedure is recommended to open the contact with the adjacent tooth and to compensate for the thickness of the matrix band. It has been demonstrated that multiple wedging (ie, inserting a wedge initially and then reapplying seating pressure several times during the course of the procedure) is more effective in opening the contact than is a single placement of a wedge.[118] In addition, the wedge can protect the rubber dam from damage and the gingival tissues from laceration, thereby reducing leakage into the operative site.[27] Alternatively, a sectional matrix band ring can be used to effectively provide tooth separation.[216] Tooth separation obtained from prewedging promotes more conservative preparation and helps protect adjacent teeth from damage during preparation (Fig 11-15). Failure to take measures to protect adjacent teeth during proximal surface preparation with rotary instruments will usually result in damage to the adjacent teeth[217] (Fig 11-16). Furthermore, this damage makes it significantly more likely that the damaged surface will require subsequent restoration.[218] Strategies for protecting adjacent proximal surfaces of adjacent teeth during preparation of a proximal area are presented in Fig 11-17.

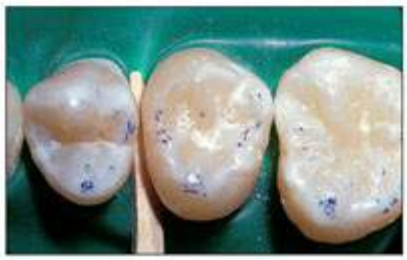


Fig 11-15 A wedge is placed before preparing the mesial surface of the maxillary second premolar; it provides tooth separation to help ensure adequate interproximal contact in the definitive restoration, and it helps prevent damage to the adjacent tooth, rubber dam, and gingival tissues.

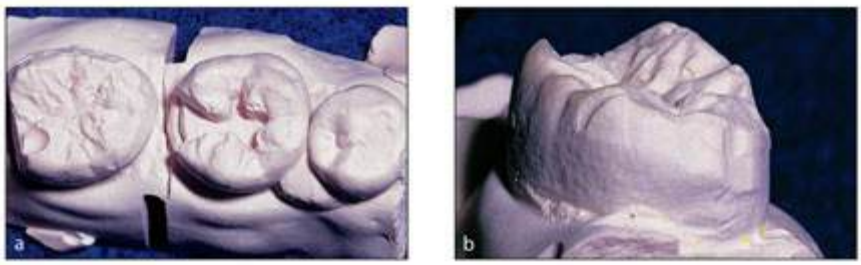


Fig 11-16 *(a)* Stone cast made following Class 2 preparation involving the distal surface of a mandibular first molar. *(b)* View of the mesial surface of the mandibular second molar shows the damage that resulted from inadequate protection during preparation of the distal surface of the first

molar.



Fig 11-17*(a)* A small piece of matrix band is placed prior to a Class 2 preparation to protect the proximal surface of the adjacent tooth. A wooden wedge has been placed to help maintain the matrix in position and to gain separation to aid in obtaining an appropriate contact in the definitive restoration. *(b)* Alternate strategies for protecting the proximal surfaces of adjacent teeth during tooth preparation of the second premolar: Interguard (Ultradent) is placed between the first and second premolars, and a circumferential metal matrix band is placed on the first molar. *(c)* The FenderWedge (Garrison Dental Solutions) is designed to provide simultaneous prewedging and protection of the adjacent tooth's proximal surface during cavity preparation. *(d)* FenderWedge in place prior to initiating preparation.

An alternative strategy for preparing the proximal surfaces of Class 2 preparations that provides protection of the adjacent proximal surface is the use of a special diamond preparation tip, half of which is "safe-sided" (ie, the diamond abrasive is only present on one side of the tip) (Figs 11-18a and 11-18b). These tips, which come in a variety of shapes and sizes (Fig 11-18c), can be used in sonic handpieces that fit a normal dental unit coupler or in a piezoelectric ultrasonic unit as might be used for scaling and root planing. It will usually be necessary to initiate the preparation with a rotary high-speed handpiece, because preparation with ultrasonic and especially sonic handpieces can be slower than with normal rotary handpieces.[219] Research has shown that pulpal temperature increase with ultrasonic preparation is not greater than that with water-cooled rotary preparation.[219] In addition, sonic preparation does not adversely affect adhesion as compared to rotary handpiece preparation with a carbide bur.[220] An example of the use of sonic preparation is shown in Fig 11-19.



Fig 11-18 *(a)* Example of a sonic preparation tip (KaVo), demonstrating the use of the diamond abrasive side to shape a proximal cavity preparation. *(b)* Example of the same sonic preparation tip, demonstrating the use of the safe, nonabrasive side to prevent damage to the adjacent tooth. *(c)* Examples of some of the many different sizes and shapes available for use in sonic or ultrasonic handpieces. Note that there are matching shapes/sizes for use on both the mesial and distal surfaces.

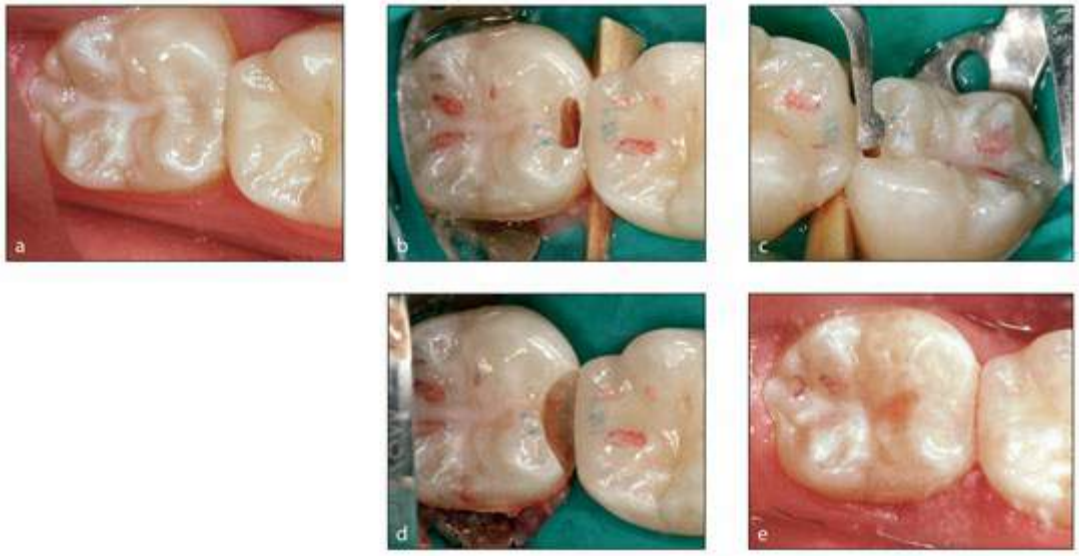


Fig 11-19 *(a)* Preoperative photograph of the mandibular left second molar. Bitewing radiographs revealed a caries lesion on the mesial proximal surface. Note that the occlusal surface had been treated with a pit and fissure sealant and was protected from caries, allowing tooth preparation to be limited to the proximal surface affected by caries. *(b)* Proximal preparation was initiated with a ¼ round bur and a rotary high-speed handpiece. Note that the proximal wall has been left intact. *(c)* A safe-sided diamond tip is used to remove the proximal wall. *(d)* Final preparation, after caries removal and refinement of the preparation with the safe-sided diamond preparation tip in a sonic handpiece. *(e)* Finished restoration.

Preparation

As a general principle, preparation should be limited to eliminating carious tooth structure and providing access for restoration placement and finishing (Figs 11-20a to 11-20c). If there are one or more areas of fissure caries lesions in the tooth, in addition to the proximal surface lesion(s), they should be treated separately, if possible, as described in the section on preventive resin restorations.

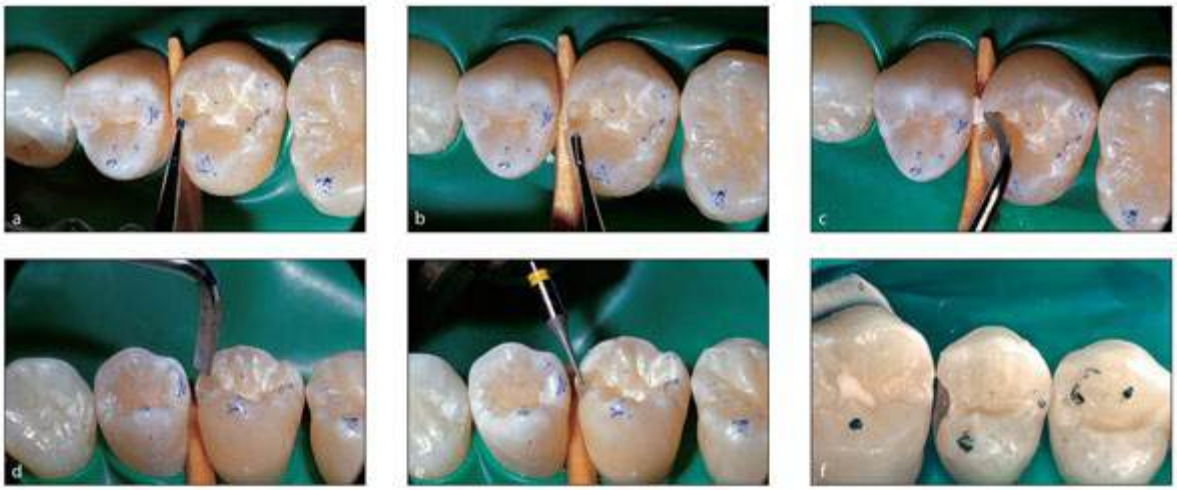


Fig 11-20*(a)* Preparation is initiated just inside the marginal ridge with a small round bur (no. ¼). *(b)* The preparation is extended with a no. 329 bur. Note that the proximal surface is left intact. *(c)* After the proximal surface is thinned, a spoon excavator is used to fracture and remove the thinned enamel. *(d)* Bevels at the cavo-surface margins of the proximal walls are placed with hand instruments. *(e)* Alternatively, fine-grit diamonds or carbide finishing burs can be used to place bevels. *(f)* When a preparation exits the external tooth surface at an obtuse angle (greater than 90 degrees), no beveling of proximal walls is necessary.

Bevel placement is a point of controversy with this preparation. When used in conjunction with adhesive agents and resin composites, bevels in enamel provide more area for acid etching and bonding. In addition, the bevel is designed to expose enamel rods transversely (cross-cut or "end-on") to achieve a more effective etching pattern. Research has indicated that etching of transversely exposed enamel rods (ends of rods) results in a bond that is significantly stronger than that attained with etching of longitudinally cut enamel rods (sides of rods).[221] Clinical research has demonstrated favorable results with the use of acid-etched, beveled preparations in Class 3 resin composite restorations.[222] The clinician must be judicious in bevel location for posterior resin composite restorations.

Following are recommendations regarding bevel placement in Class 2 preparations for resin composite restorations.

Facial and lingual proximal margins

Because enamel rods exit the tooth at approximately right angles to the external tooth surface, it is necessary for the cavity preparation to form an obtuse angle (greater than 90 degrees) with the external tooth surface to expose the ends of the enamel rods. If the external cavosurface margin forms a right angle with the tooth surface, conservative bevels (0.5 mm) should be placed at an approximately 45-degree angle to the surface, on the facial and lingual cavosurface margins of the proximal box

preparation (Figs 11-20d and 11-20e). This will achieve the benefits of beveling as well as aid in placing the margins in a more accessible location for finishing and polishing. Research has demonstrated that bevels on these margins significantly reduce marginal leakage.[223,224] If the preparation exits the tooth at an obtuse angle, no further beveling of the proximal walls is necessary (Fig 11-20f).

Gingival margins

The decision to place a gingival margin bevel requires clinical judgment. The gingival margin should be beveled only if the margin is in enamel well away from the cementoenamel junction and an adequate band of enamel remains (Fig 11-21; see also Fig 11-2c). When sufficient dentin-supported enamel remains for adequate bevel placement, resin composite adaptation is enhanced.[225] As the preparation nears the cementoenamel junction, the enamel layer is thinner than in other regions of the crown, and beveling the preparation increases the potential for removing the little enamel that remains. Because of the presence of prismless enamel in this region, acid etching is often less effective.[226] When a cavity preparation approaches within approximately 1 mm of the cementoenamel junction, adhesion is essentially no better than bonding to dentin[227,228] (Fig 11-22).

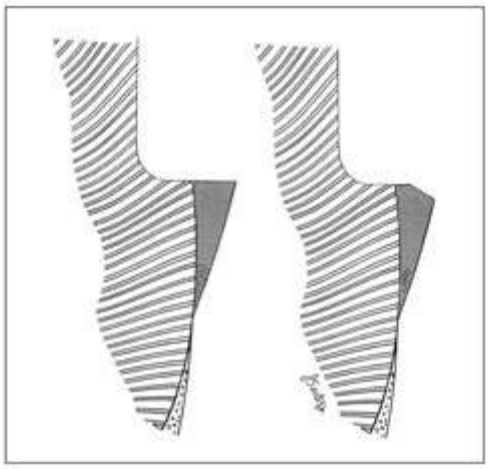


Fig 11-21 When enamel is adequate at the gingival margin, a bevel is placed to enhance resin composite adaptation and seal.



Fig 11-22 When adequate enamel remains at the gingival margin, a cavosurface margin bevel is

placed to expose the ends of the enamel rods for etching. *(a)* Appropriate proximal cavosurface margin bevels when the enamel is well above the cementoenamel junction. Gingival as well as facial and lingual vertical cavosurface margins are beveled. *(b and c)* As the gingival margin approaches 1.0 to 1.5 mm from the cementoenamel junction *(b)*, or when it is apical to the cementoenamel junction *(c)*, no gingival bevel is placed. Note that the facial and lingual vertical cavosurface margins are still beveled.

Use of an *inverse* or *internal bevel*, leaving enamel that is not supported by dentin at the gingival cavosurface margin (Fig 11-23), has been shown to significantly reduce microleakage as compared to a butt margin[229] (see Fig 11-23b) and would be preferable to placing the gingival margin on or near the cementoenamel junction. This type of marginal configuration should not be created intentionally with a bur, but if a lip of unsupported enamel remains after removal of demineralized dentin, it should be configured to an inverse bevel rather than planing the unsupported enamel off to form a butt margin in cementum or dentin.

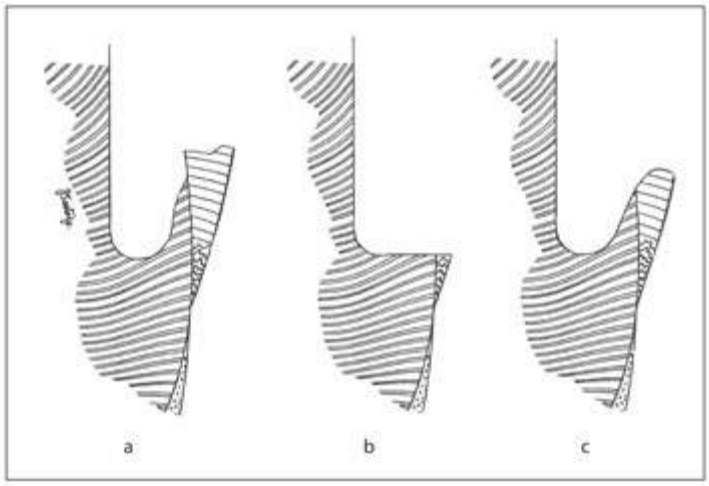


Fig 11-23*(a)* Excavation of a proximal caries lesion will sometimes result in a more gingival extent of the preparation in dentin than in enamel. *(b)* If the preparation is extended straight out to the cavosurface margin, remaining enamel for bonding is compromised. *(c)* Refining the preparation with a low-speed round bur or a sharp hand instrument to eliminate very thin enamel and prepare an inverse bevel will expose enamel rods for etching on their internal ends and secure better adhesion to the gingival margin. Assuming that removal of carious dentin created the situation shown in *a*, the marginal configuration shown in *c* is preferable to that in *b*.

Occlusal margins

The use of occlusal cavosurface margin bevels is not indicated. Some clinicians have advocated the use of bevels on occlusal cavosurface margins to maximize the exposure of end-cut enamel rods.[226] However, it has been noted that a normal preparation in the occlusal surface will result in end-cut enamel rods because of the orientation of the enamel rods in cuspal inclines[163] (Fig 11-24). Avoidance of

bevels on the occlusal surface prevents the loss of sound tooth structure, decreases the surface area of the definitive restoration, lessens the chance of occlusal contact on the restoration, eliminates a thin area of resin composite that would be more susceptible to fracture and wear, and presents a well-demarcated marginal periphery to which resin composite can be finished more precisely.[14,230,231]

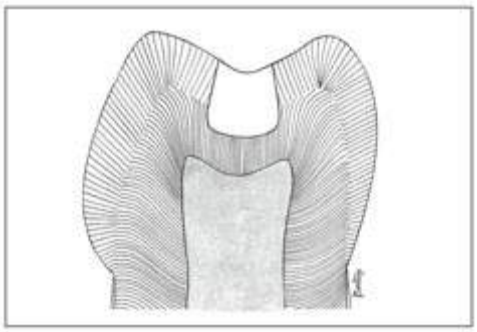


Fig 11-24 The enamel rods on the occlusal surface are oriented in such a way that the ends are exposed without beveling.

Placement of occlusal bevels has demonstrated no benefit to the longevity of Class 2 resin composite restorations.[30] A clinical study demonstrated that posterior resin composite restorations placed in cavities with beveled occlusal margins had significantly greater wear than those placed with occlusal butt margins.[213] A significant factor predicting the survivability of posterior resin composite restorations is the proportion of the occlusal surface restored[16,20,21,30]; this factor is increased by occlusal beveling. Therefore, occlusal cavosurface margin bevels should be avoided (Fig 11-25).



Fig 11-25 Finished preparation. The tooth has been prepared only in those areas where carious dentin was present. The preparation has not penetrated to dentin in other areas of the occlusal surface. Proximal facial, lingual, and gingival cavosur-face margins have been beveled; occlusal cavosurface margins have not been beveled.

It should be noted that occlusal enamel should not be left unsupported by dentin, particularly in an area of occlusal stress. Research has shown that unsupported

occlusal enamel, even if the lost dentin has been replaced with glass ionomer, RMGI, or bonded composite, is significantly weaker than enamel supported by dentin.[232]

Use of cavity liners

If used, calcium hydroxide liners should be limited to those areas of the preparation that are believed to be very close to the pulp, where there is the possibility of a minute pulpal exposure.[96] Placement of a calcium hydroxide liner over an extensive area of dentin provides no benefit to the pulp and decreases the surface area of dentin available for adhesion. Dissolution of the liner during acid etching can interfere with a sound bond to enamel and dentin.[233] If the preparation is conservative in size, no liner is required in addition to the adhesive agent. In deeper preparations and those in which the gingival margin approaches or extends beyond the cementoenamel junction, a glass-ionomer liner may be beneficial. Pulpal considerations are discussed in depth in chapter 6.

Glass-ionomer liners are reported to offer a number of potential advantages when used under posterior resin composite restorations. Glass-ionomer materials bond to both tooth structure and overlying resin composite,[234] and they introduce less polymerization stress into tooth structure than does resin composite.[79] Glass ionomer releases fluoride into adjacent tooth structure,[235] which may be advantageous because of the tendency for secondary caries lesions to occur adjacent to posterior resin composite restorations.[14] Use of a glass-ionomer liner has been demonstrated to improve marginal integrity[236] and decrease marginal leakage.[169,237,238] Less bulk of resin composite material is required to fill the preparation, reducing the amount of polymerization shrinkage[239] and improving marginal adaptation.[68] Glass-ionomer liners can reinforce the preparation walls by adhering to dentin and minimizing cuspal deformation under load.[239] Glass-ionomer liners also reduce the rise in pulpal temperature associated with application of the curing light during incremental insertion procedures.[240] With improvements in dentin adhesives, the use of glass ionomer under posterior resin composite restorations has been greatly reduced in recent years. However, use of a glass-ionomer liner on dentin cavity surfaces has been shown to significantly reduce postoperative sensitivity compared to use of a dentin adhesive alone.[241] Therefore, clinicians should consider use of a glass-ionomer liner even if an adequate band of enamel surrounds the entire preparation.

Bonded-base technique

If the gingival margin of a Class 2 preparation is in enamel but within 1 mm of the cementoenamel junction, or if it is in dentin, and an alternative restorative material is not appropriate (eg, amalgam), an RMGI restorative material should be placed as the initial increment in the proximal box. This technique, known as the *bonded-base* o r *open sandwich technique*, has demonstrated a number of advantages when compared with use of an adhesive agent alone. There is considerable in vitro evidence that the bonded-base technique results in improved marginal adaptation and a reduction in marginal leakage.[242–249] Additionally, glass ionomers have demonstrated good antibacterial activity against microorganisms associated with dental caries,[250] as well as reduced demineralization adjacent to dentin margins.[251] Most important, the bonded-base technique has demonstrated good clinical performance. The first evaluation of the bonded-base technique used a conventional restorative glass ionomer as the initial increment in the proximal box. Unfortunately, this technique showed poor clinical longevity.[252] However, use of an RMGI restorative material for the initial increment in the proximal box has proven to be a viable technique. One clinical study showed that RMGI provided significantly better cervical margin adaptation in Class 2 cavity preparations than did resin composite.[253] Several studies have shown that the bonded-base Class 2 restoration performs comparably with resin composite restorations using adhesive systems alone in most clinical parameters.[152–155] In addition, the bonded-base technique has demonstrated some important clinical advantages versus bonded Class 2 composite, including reduced postoperative sensitivity[155] and reduced in vivo demineralization adjacent to the gingival margin.[152] For these reasons, the bonded-base restoration should be considered whenever a posterior resin composite restoration will include a gingival margin in close proximity or apical to the cementoenamel junction.

The technique for accomplishing a bonded-base restoration is demonstrated in Fig 11-26. After completion of the preparation, the matrix is applied, and a wedge is placed. The gingival portion of the proximal box is treated with the RMGI conditioner. The RMGI is mixed, transferred to a light-protected Centrix syringe tip, and injected into the gingival aspect of the proximal box. Because glass ionomer does not have the same level of wear resistance as composite,[254,255] this increment of RMGI should remain apical to the proximal contact (Fig 11-26c). The surface of the RMGI increment is smoothed and light cured. The entire cavity preparation, including the RMGI in the gingival portion of the proximal box, is etched, and the adhesive system is applied according to the manufacturer's instructions. The

preparation is filled and finished as described in the subsequent sections of this chapter.

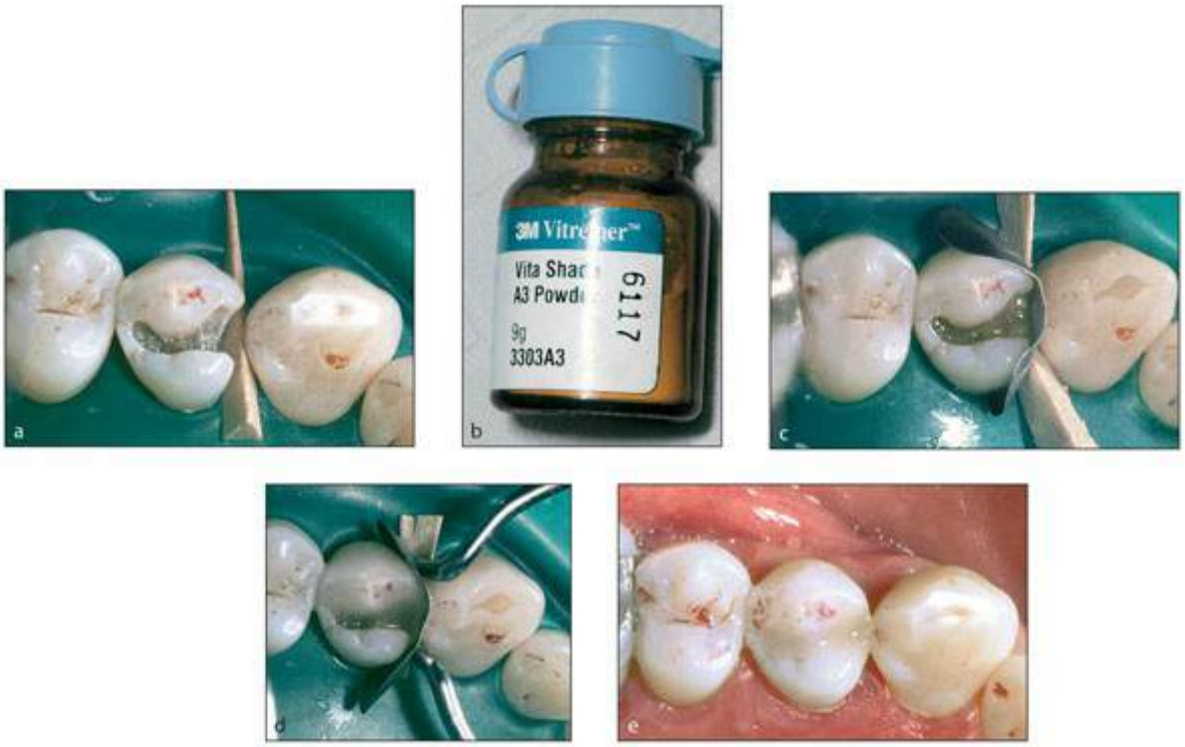


Fig 11-26*(a)* Following caries excavation, the gingival margin of the proximal box extends below the cementoenamel junction onto dentin. Note that the gingival margin extends to the fluted area of the root. *(b)* To perform the bonded-base technique, an RMGI restorative (versus a lining glass ionomer) is used. In this case, Vitremer (3M ESPE) was used. *(c)* Following treatment of the gingival portion of the proximal box with the RMGI conditioner, the RMGI restorative is mixed, placed into a Centrix syringe tip, injected into the gingival portion of the proximal box, smoothed, and cured. The RMGI should remain apical to the proximal contact. Note that two wedges have been used to contour the gingival margin of the matrix into the fluted area of the root. *(d)* Following etching and application of the adhesive system to the RMGI and remaining aspects of the cavity preparation, the preparation is filled using an incremental placement technique. *(e)* The completed restoration.

Adhesive system application

The adhesive system of the dentist's choice should be applied. Chapter 9 provides detailed information on adhesive system selection and application.

If the preparation is etched and the bonding resin is placed before application of the matrix, visualization and access to all areas of the preparation are better, and it is easier to brush-thin the adhesive and avoid pooling. A small piece of matrix band material, used to protect the adjacent proximal surface, may be kept in place. However, the wedge either should not be replaced at this time or should be placed between the matrix material and the preparation. This gives the resin adhesive

access to all areas of the preparation and provides an escape for the resin adhesive at the margin to prevent it from pooling. Placement of a matrix after adhesive application sometimes results in contamination of the preparation with blood or saliva, so the operator may prefer to place the matrix and wedge prior to etching and application of the adhesive. If so, special care must be taken to ensure the absence of pooling of the resin adjacent to the matrix. Enamel and dentin adhesives are discussed at length in chapter 9.

Matrix application

Several useful matrices are available, including the clear plastic matrix, the ultrathin (0.001-inch) Tofflemire metal matrix (Fig 11-27a), the thin (0.0015-inch) sectional matrix, and the Tofflemire metal matrix with photoetch-thinned (0.0005-inch) contact areas.

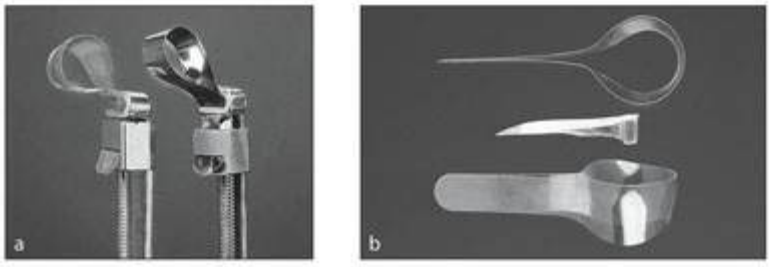


Fig 11-27 Two common matrices for Class 2 resin composite restorations. *(a)* Clear and metal matrix bands are shown with Tofflemire retainers. *(b)* The clear matrix is usually used with a light-reflecting wedge.

The clear matrix can be used in conjunction with a light-reflecting wedge (Fig 11-27b) and offers the advantage of allowing penetration of the curing light from multiple directions. This allows the clinician to cure the increments of resin composite from the proximal and gingival directions, rather than from the occlusal aspect only, to ensure adequate polymerization of each increment. In addition, it has been reported that this technique allows more favorable direction of the polymerization shrinkage. One study showed enhanced gingival margin adaptation using this technique when the proximal box was prepared in enamel,[256] although another study failed to show a similar benefit when the gingival margin was in dentin.[228] However, the clear matrix is thicker than the thinnest metal matrices, and its lack of rigidity makes placement through tight interproximal contacts difficult.[231] In addition, the rigidity and smoothness of the plastic, light-reflecting wedge makes it less effective than a wooden wedge in gaining the slight tooth separation needed to ensure adequate interproximal contact. Methods to compensate for this lack of

separation are described later in this chapter. However, because of these drawbacks, most clinicians prefer metal matrices and wooden wedges.

Tight interproximal contacts are more easily developed with the ultrathin metal matrices than with the clear matrices because they are easier to place, maintain their shape better, and can be burnished against the adjacent tooth. One disadvantage of a metal matrix that wraps around the facial and lingual surfaces of the tooth is that increments must be initially cured only from the occlusal aspect. After removal of the matrix, the proximal resin composite may be further polymerized from the facial and lingual aspects. To avoid flat proximal-surface contours, precontoured matrices should be used or the metal matrices should be shaped or contoured by burnishing before or after they are placed. The marginal ridge strength obtained with a contoured sectional matrix is significantly greater than that obtained with a straight, circumferential band.[257,258]

Other devices that are helpful in developing adequate interproximal contacts are the sectional matrix systems, used with metal rings with springlike properties (Fig 11-28; see also Figs 11-30g and 11-30i). After the sectional matrix and wooden wedge are placed, the ring is placed using a rubber dam clamp or similar forceps so that the vertical points of the ring are positioned in the facial and lingual embrasures adjacent to the box preparation. The ring holds the ends of the sectional matrix tightly against the tooth and exerts a continuous separating force between the teeth. The matrix should be burnished gently against the adjacent proximal contact (Fig 11-29). The sectional matrices in these systems are typically made of "dead soft" metal. Heavy burnishing will cause grooves to be formed in the matrix that will be replicated in the restoration. This makes for a rough, irregular contact that can tear and shred floss when the patient performs oral hygiene measures, so only light burnishing should be used.

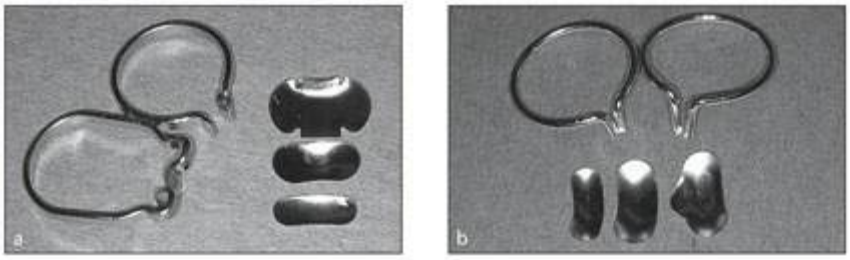


Fig 11-28 *(a)* Palodent sectional matrix system (Dentsply/Caulk). *(b)* Composi-Tight sectional matrix system (Garrison Dental Solutions).



Fig 11-29 Sectional matrix and Composi-Tight ring in place. The matrix is burnished in the contact area to enhance proximal contour and contact.

These "ring" sectional matrix systems have a number of advantages: They provide tooth separation to ensure good interproximal contact; they provide better proximal contours for posterior resin composite restorations than traditional matrices; and they simplify matrix placement for single proximal-surface restorations as compared to a circumferential band.[259,260] In addition, these systems provide a tighter, longer-lasting contact in resin composite than does a standard matrix in a Tofflemire retainer.[257,261,262] It should be recognized that the ring provides progressive tooth separation, so if it is left in place for a long period of time, excess separation can occur, resulting in a very tight contact.

The placement of the ring relative to the prepared tooth and the wedge should vary based on the location of the proximal margins and the type of ring. Proper placement of the ring depends on facial and lingual extensions of the proximal box and the size and shape of the tines of the particular ring being used. Rings with wider tines (eg, the Palodent system, Dentsply/ Caulk) tend to limit the ability to place the ring in different locations relative to the wedge, whereas rings with narrower tines (eg, the Composi-Tight system, Garrison Dental Solutions) provide more flexibility in this regard. If the facial and lingual proximal extensions of the proximal box do not extend significantly onto the facial or lingual surfaces of the tooth, it is possible to place the ring with the tines occlusal to the wedge (Figs 11-30a and 11-30b) or between the wedge and enamel adjacent to the proximal surface being restored. However, if one or both of the proximal walls reach the facial or lingual surfaces of the tooth, placement of the tines of the ring in these locations may cause the matrix to be deformed (Fig 11-30c). In this case, the tines may be placed between the wooden wedge and the proximal surface of the adjacent tooth. The ring will have enough tension to separate the teeth adequately and to cause the wedge to wrap slightly around the tooth, providing a tight gingival seal and wrapping of the sectional matrix around the tooth to form the proper proximal contour (Fig 11-30d). Plastic wedges have been developed that allow additional wrapping of the wedge and matrix when the ring is placed between the wedge and the adjacent tooth surface (Figs 11-30e

and 11-30f). Newer rings have been designed to help overcome some of the concerns regarding the placement of the ring relative to the position of the wedge or the preparation walls. Some rings incorporate an inverted "V" shape, which allows the ring to straddle the wedge, provide tooth separation, and contour the matrix around the facial and lingual contours of the tooth (Figs 11-30g to 11-30k).



Fig 11-30 *(a)* Conservative proximal preparation allows the ring for a sectional matrix system to be placed above (coronal to) the wedge. This placement provides a separating action and contours the matrix to provide correct proximal contours. *(b)* Facial view showing placement of the ring. *(c)* In this preparation, the lingual proximal wall extends onto the lingual surface. The ring has been placed above (coronal to) the wedge. However, the lingual tine has caused the matrix to partially collapse into the preparation, resulting in the matrix band covering a portion of the gingival floor of the preparation. *(d)* The ring has been respositioned between the wooden wedge and the adjacent tooth. This prevents the tines from collapsing the matrix into the preparation. It also causes the wedge to bend, enhancing the gingival seal of the matrix and wrapping the matrix around the tooth to develop the appropriate proximal contour. *(e)* Plastic wedges are more flexible than wooden wedges and may be more useful when the ring must be placed between the wedge and the adjacent tooth (Wedge Wands, Garrison Dental Solutions). *(f)* The ring placed between the plastic wedge and the adjacent tooth. Note the pronounced wrapping of the wedge at the gingival aspect of the preparation. *(g)* Underside of an original sectional matrix ring *(left)* next to a sectional matrix ring (Garrison Dental Solutions) that incorporates a "V" shape to allow the tines to straddle

the wedge *(right)*. *(h)* The matrix ring in position. Note the "bumpers" on the tines that aid in adapting the matrix to the facial and lingual contours of the tooth. *(i)* Original "V-wedge" designed with tines to sit on top of an inserted interproximal wedge (Triodent). *(j)* V-wedge in place. Note that the placement of the tines is not affected by the wedge and that the matrix is contoured around the facial and lingual contours. *(k)* Current iteration of the V-wedge (Triodent).

Yet another type of sectional matrix involves a short piece of thin, stainless steel matrix material that is contoured only occlusogingivally and does not surround the tooth at all (Fig 11-31). This allows some light curing from facial and lingual aspects. The contoured matrix is secured with a wooden wedge and lies passively against the adjacent tooth surface or is held there with an instrument during curing of the first increment. That increment then holds the matrix against the adjacent tooth during placement of succeeding increments. When this technique is used, resin composite in facial and lingual embrasures must be contoured or sculpted with a thin-bladed instrument, such as an interproximal carver (IPC), prior to curing.



Fig 11-31 A passive sectional matrix is placed and wedged. Resin composite will be sculpted in the facial and lingual embrasures with a thin instrument such as an IPC.

Resin composite placement: Incremental technique

VLC resin composite should be placed in successive, laminated increments to ensure proper curing and prevent excessive polymerization shrinkage stress.[13] Incremental curing decreases the effects of polymerization shrinkage, enhances marginal adaptation, decreases gap formation, reduces marginal leakage, decreases cuspal deformation, makes the cusps more resistant to subsequent fracture, improves bond strength to cavity walls, and decreases postoperative sensitivity.[66,68,256,263–266] One of the greatest benefits to the incremental fill technique may be its effect on cavity configuration, or *C-factor*.

The C-factor is the ratio of bonded to unbonded restoration surface areas and has been shown to have a profound impact on polymerization shrinkage stress (see chapter 9). As the C-factor increases, that is, as the surface area of resin composite bonded to cavity walls increases relative to the surface area of unbonded resin composite, shrinkage stresses increase dramatically.[75,267,268] With incremental

placement and curing of resin composite, the C-factor of each increment is reduced compared with bulk placement and curing.[269] As the C-factor decreases, bond strength increases.[270] The end result is that the incrementally placed and cured restoration is bonded better to the cavity walls than if the preparation had been filled and the resin composite material cured in bulk.

First increment

Some general guidelines should be followed for the placement of the resin composite. Proper handling of the bonding system and resin composite at the gingival margin is critical because of the tendency for microleakage to occur in that area.[238] A clinical study that assessed gingival margin quality in Class 2 resin composite restorations showed that only 27% were satisfactory.[271] Therefore, techniques must be used to enhance the bond and reduce the adverse effects of polymerization shrinkage. First, an increment no thicker than 1 mm is placed against the gingival wall.[272] A thin first layer will ensure proper light irradiation throughout the increment.

If a clear matrix and light-reflecting wedge are being used, the initial curing should be directed through the flat end of the wedge. The scientific literature reveals that the amount of light that a reflecting wedge will transmit varies. One study indicated that 90% to 95% of the incident light is transmitted,[256] while another showed that it could be as low as 66%.[273] Because of the possible attenuation of light through the wedge, exposure time should be increased by 50% to ensure adequate polymerization. It has been suggested that the light-reflecting wedge will direct the curing light to the gingival margin of the restoration and draw the polymerization shrinkage toward that margin. When the gingival margin is on enamel, this method has been shown to result in better marginal adaptation than curing from an occlusal direction.[68,256,274,275] However, when the gingival margin is on dentin, this technique has failed to reduce gingival margin microleakage compared with other techniques.[228]

Because most plastic wedges are rigid and smooth, they may slip out of proper position easily and may not maintain the pressure necessary to ensure proper adaptation of the gingival aspect of the matrix band and separation from the adjacent tooth. Two suggestions may help to overcome this problem. After the plastic wedge is positioned, a wooden wedge may be inserted beside it on the side away from the tooth being restored. Alternatively, the plastic wedge can be maintained in its proper position by applying pressure to its flat end with the light-curing tip during curing of

the initial increment (Fig 11-32). After this is completed, the plastic wedge has accomplished its purpose and may be replaced with a wooden wedge for succeeding increments.



Fig 11-32 When a clear matrix and reflecting wedge are used, the initial cure for the gingival resin composite increment is through the wedge. The end of the light guide can be used to maintain pressure on the smooth plastic wedge during initial polymerization to help ensure complete insertion of the wedge. Because of loss of irradiance when curing through a wedge, cure times should be increased by one-half.

If a metal matrix that surrounds the tooth has been chosen, all increments must be cured from the occlusal aspect. The tip of the light should be positioned as close as possible to the resin being cured.[6] The output of curing lights diminishes considerably with distance, with 50% of light lost just 3 to 6 mm from the light tip.[276] After the metal matrix is removed, all proximal areas of the restoration should receive additional curing with the light to maximize restoration cure.[277]

Resin composites that are marketed for posterior use vary widely in their viscosity.[278] This can have an impact on adaptation of resin composite to the walls of a cavity preparation.[279] Adaptation of resin composite to cavity walls can have a dramatic effect on the bond strength; as adaptation worsens and voids increase, the bond decreases significantly.[280] Thicker-consistency resin composites have significantly increased cavity-wall voids compared with medium- or thinner-viscosity materials.[278] Resin composites that are supplied in preloaded resin composite tips or ampules tend to have a higher viscosity than do composites that are supplied in syringes.[278] Placement technique can also determine how well the resin composite adapts to the cavity-preparation walls. Use of Centrix placement tips (Fig 11-33) for resin composite decreases the viscosity of the material[278] and significantly decreases voids adjacent to the preparation walls compared with either smearing the material into place with a plastic instrument or “condensing” it.[279]

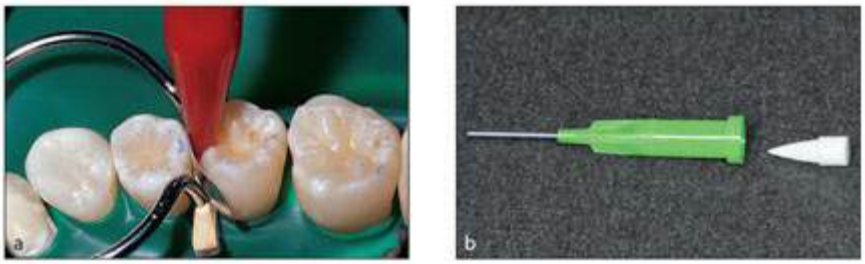


Fig 11-33 *(a)* Previously warmed resin composite is syringed into the preparation via a Centrix placement tip to maximize adaptation to the cavity walls. *(b)* Another Centrix syringe tip that can be used in conjunction with some warm resin composites has a metal tip that can be easily formed to various angles.

Injecting a heavily filled composite through a narrow placement tip can be difficult, if not impossible. A technique that will further enhance the flow of the resin composite into a cavity preparation is to use resin composite that has been warmed prior to injection.[281] The required amount of resin composite from either a syringe or a premanufactured compule tip is transferred into a Centrix syringe tip. The tip and composite can be warmed in a water bath; however, the tip should be sealed in a small plastic bag prior to immersion to protect the resin composite material from moisture. A more convenient means for warming composite is to use a commercial composite warmer (Calset, AdDent) to reduce the viscosity and improve flow.[282] The temperature to which the composite is warmed is based somewhat on individual preference but typically will be in a range of 140°F to 155°F (60°C to 68°C). This material can then be syringed into place more easily, and the lowered viscosity enhances resin composite adaptation to the cavity walls (see Fig 11-33a).

In addition to the above advantages for injecting warmed resin composite (ie, reduced viscosity and improved resin composite flow) other advantages have been demonstrated. Use of warmed resin composite leads to reduced marginal leakage in Class 2 resin composite restorations compared with using either room-temperature resin composite or flowable composite.[283,284] Warmed resin composite exhibits improved cure, as compared to room-temperature resin composite.[285,286] However, the rapid rate at which the warmed composite loses heat once it is placed into contact with the tooth likely minimizes the enhanced cure, and it does not therefore cause a significant increase in shrinkage.[287] Because most temperature increase is due to reaction kinetics and heat from the curing unit, the warmed resin composite adds very little temperature increase to that which already occurs.[285] Resin composite does not polymerize in the warming unit, even if kept at 130°F (55°C) for 4 hours or 158°F (70°C) for 15 minutes.[285] Finally, prewarming of the resin composite does not adversely affect the material's strength.[288]

Flowable resin composites

Another method that has been suggested is the use of low-viscosity, or *flowable*, resin composites for the first increment of a proximal box or pulpal floor.[289] The rationale is that these materials flow more readily than standard hybrid formulations and will therefore easily and thoroughly adapt to all areas of the cavity preparation. Also, because of their lower filler content and reduced elastic modulus, it is theorized that these materials could act as "stress relievers" to absorb forces of polymerization shrinkage or cyclic loading.[290] However, the efficacy of this method has not been demonstrated.[290]

There are a number of problems associated with these materials. Because of their higher resin content,[291] flowable resin composites demonstrate up to three times greater polymerization shrinkage than do standard hybrid resin composite formulations.[291,292] This generates significantly greater polymerization shrinkage forces that surpass any benefit that might be derived from the lower elastic modulus.[293] This adversely impacts the adhesion of the resin composite to the cavity preparation, as higher polymerization shrinkage[294] and polymerization shrinkage stress[270] have been shown to significantly decrease bond strength. To address this issue, newer flowable dental composites with relatively high filler content have been developed that have lower shrinkage stress–generating characteristics due to specific modifications of their chemical formulations.

The use of a cured increment of a flowable resin composite in conjunction with Class 2 resin composite restorations has shown mixed results in studies of marginal leakage. Some studies have shown improved marginal seal,[295,296] and others have indicated worsened gingival seal.[297,298] However, the majority of studies show no benefit from using a cured portion of flowable composite as the initial increment in placement of a Class 2 resin composite restoration.[248,299–303] In addition, because of their lower filler load, many flowable resin composites have inferior mechanical properties compared with traditional resin composites.[290] This leads to concern that occlusal forces may introduce increased deflection of the overlying hybrid resin composite, which has a higher modulus of elasticity, due to the inability of the flowable resin composite to provide adequate support.[290] In fact, the use of a flowable resin liner in conjunction with a high-viscosity (packable) resin composite has been shown to reduce the strength of the polymerized packable material.[304]

The reduced viscosity of flowable resin composites improves adaptation to the

preparation but may also result in undesirable effects. One study found that use of a flowable composite always led to increased incidence of gingival margin overhangs in beveled Class 2 cavity preparations.[305]

An additional concern with flowable resin composites pertains to their radiopacity. Detection of voids and recurrent caries lesions is maximized when a restorative material has the same radiopacity as, or slightly greater radiopacity than, enamel.[48–50] Many flowable resin composites have not met the standard of being at least as radiopaque as enamel.[51,52,306–308]

Clinical trials using flowable resin composite for the initial increments in posterior resin composite restorations have not demonstrated any difference in either postoperative sensitivity[309,310] or overall clinical performance[310,311] compared with the use of heavily filled resin composite alone.

One suggestion that may hold promise for the use of flowable composites in conjunction with posterior resin composite restorations has been called the *snowplow technique*.[312] In this technique, an initial thin increment of flowable composite is placed over the gingival and/or pulpal floors of the cavity preparation. This layer is not cured at this stage, but rather an initial increment of heavily filled restorative resin composite is syringed or pushed into the unset flowable resin composite. Most of the flowable resin composite is displaced by the restorative composite and is subsequently removed from the cavity preparation with a hand instrument, microbrush, or bristle brush. As a result, most of the flowable composite, and therefore its potentially disadvantageous characteristics, is not present in the cavity preparation. Instead, there is only a small amount of flowable resin composite remaining in those areas of the cavity in which the higher-viscosity resin composite did not completely adapt to the preparation and that otherwise may have been void of restorative material. The combined increment of flowable resin composite and restorative resin composite is then cured. This technique has demonstrated significantly reduced void formation compared with placement of restorative composite alone.[312] It has also shown significantly decreased gingival margin leakage in Class 2 resin composite restorations when compared with use of a restorative resin composite alone or with placement of a cured increment of flowable resin composite prior to restorative resin composite placement.[248] It should be noted, however, that no clinical studies with this technique have been reported.

The physical and mechanical properties of flowable composite resins have improved considerably since their introduction. Some flowable composite resins

demonstrate properties comparable with those of hybrid restorative formulations.[313] Recently, a new composite resin and delivery system (Sonic-Fill, Kerr) has been introduced that uses a special handpiece to deliver sonic energy to a relatively stiff composite resin to dispense it into a cavity preparation quickly and easily. This method, like the warmed composite method, allows one to make a highly filled composite resin flow much more readily to fill and adapt to a cavity preparation, without compromising its mechanical properties. New "self-adhesive" flowable composites have also been developed; these are intended to be used alone, minus the additional step of bonding with a separate adhesive (Vertise Flow, Kerr; Fusio Liquid, Pentron). It is too soon to know if these materials will demonstrate adequate clinical performance. However, it is obvious that as flowable composite resins continue to improve, their use in posterior restorations will undoubtedly increase.

Additional increments

Subsequent increments should be placed in thicknesses no greater than 2 mm. An oblique layering technique should be used whenever access allows (Fig 11-34). An oblique layering technique is preferred because it leads to higher bond strength compared with either the use of horizontal increments or bulk placement.[265,314] In addition, incremental techniques in which the facial and lingual walls are linked by the composite increment during curing tend to show greater cuspal deformation, particularly when the final, occlusal composite increment engages both the facial and lingual cavity walls.[315,316] The restoration should be cured from the facial and lingual aspects after removal of the matrix (Fig 11-35). If a clear matrix is employed, the oblique technique (see Fig 11-34a) should be used, and additional curing from the facial and lingual aspects can be accomplished through the matrix to ensure a thorough cure.

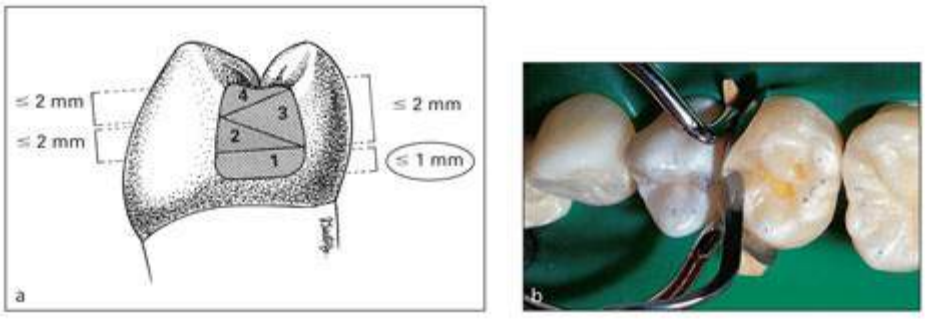


Fig 11-34 *(a)* If a metal matrix is used with a light-curing resin composite, a minimal (1 mm) gingival increment is placed and cured. The material is then layered in alternating oblique increments. No increment exceeds 2 mm in thickness. To minimize cuspal deformation and polymerization stress, an oblique increment should not contact both facial and lingual cavity walls. This placement technique can also be used with a clear matrix and reflecting wedge. *(b)* After the

initial increment is cured, the next increment is syringed into place and "ramped" obliquely with a plastic instrument or IPC. Here, the Composi-Tight matrix and ring are used, and an oblique increment is shaped with an IPC.

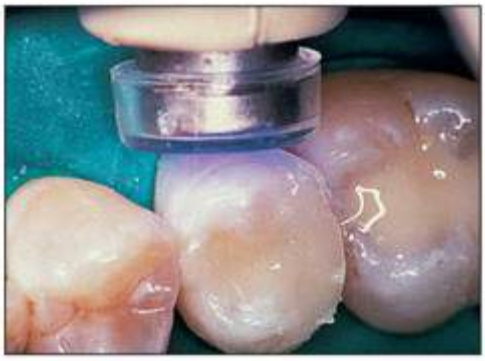


Fig 11-35 If a surrounding metal matrix was used, the restoration should be cured from the facial and lingual aspects after matrix removal.

When the proximal boxes have been filled and the resin composite polymerized, the occlusal channel, if present, is filled and cured incrementally. Alternatively, after the proximal box has been filled to the level of the pulpal floor, the proximal box and occlusal preparations can be incrementally filled and cured simultaneously. With the exception of the initial increment in the gingival aspect of the proximal box, subsequent resin composite increments should not contact both the facial and lingual preparation walls simultaneously; this is to minimize polymerization shrinkage stress[317] and cuspal deformation.[315] Figure 11-36 presents the placement of posterior resin composite restorations using a clear matrix and light-reflecting wedges.



Fig 11-36 *(a)* Preoperatively, the occlusal contacts are marked and a rubber dam is placed. *(b)* Conservative mesio-occlusal and disto-occlusal preparations have been made in the mandibular left

second premolar. Wooden wedges are in place for prewedging. *(c)* A clear matrix and reflecting wedges in place. *(d)* When the resin composite restorations are completed, a surface sealer is placed.

An alternative to the layering techniques is the use of a conical light-curing tip (Fig 11-37). The proximal box is filled with composite to just gingival to the contact area, and the conical tip is wedged into the resin composite. The cone is used to apply pressure to the matrix band and push it against the adjacent tooth during curing. Subsequent increments restore the cone-shaped gap formed by the tip. This technique is designed to ensure adequate interproximal contact and to minimize the thickness of resin composite that the light must penetrate. While the technique has not become particularly popular and is relatively untested, study results have shown some benefits, including formation of fewer marginal gaps than in the more traditional incremental techniques,[318] reduced cusp deflection,[319] improved hardness, and decreased porosity.[320]

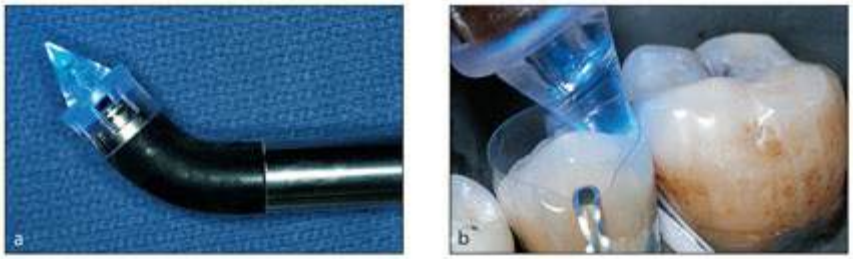


Fig 11-37 A conical light tip *(a)* can be placed into uncured resin composite increment *(b)* to provide curing while also pushing the matrix against the adjacent tooth to enhance proximal contact.

A similar technique to help establish interproximal contact is to place the first increment into the proximal box, but instead of pushing the matrix with the conical tip, it is held against the adjacent tooth during the polymerization with a plastic instrument, condenser, or similar instrument. That increment, after it is hardened, will then hold the matrix against the tooth as successive increments are placed.

Some manufacturers have devised instruments that are precontoured to provide a wedging effect of a matrix band against the adjacent tooth. These instruments are designed to ensure good contact while maintaining proper occlusogingival contours (Fig 11-38).



Fig 11-38 Examples of composite contact-forming instruments. *(a)* ContactPro (CEJ Dental). *(b)* TriMax (AdDent). *(c)* Perform Proximal Contact Instrument (Garrison Dental Solutions). *(d)* Preoperative photograph of a disto-occlusal amalgam restoration in a mandibular right second premolar to be removed due to a recurrent caries lesion. *(e)* Completed preparation with a matrix and a wedge in place. *(f)* An initial increment of composite has been placed into the proximal portion of the cavity preparation. The ContactPro has been inserted into the unpolymerized composite and rotated and wedged against the mandibular right first molar. The composite will be cured while the ContactPro is held in this position. *(g)* After removal of the ContactPro. Note that the cured composite between the tines of the contact-forming instrument is maintaining a definite interproximal contact. *(h)* Definitve restoration with appropriate proximal contact.

The use of balls of prepolymerized resin composite has also been suggested to aid in establishing interproximal contact. The normal incremental technique is used until the proximal box is filled to just gingival to the proximal contact. A small, slightly flattened ball of resin composite is precured on the tip of an instrument (eg, a Hollenback no. ½ carver). An additional increment of uncured resin composite is placed into the proximal box. The precured ball is pushed into this increment to wedge the matrix tightly against the adjacent tooth; then the resin composite is cured[321] (Fig 11-39).



Fig 11-39 Use of a precured resin composite ball to establish proximal contact. A small amount of composite is placed on the tip of an instrument (such as the Hollenback no. ½ carver) and cured. It is then pushed into uncured resin composite material in the proximal box. While the precured ball is wedged tightly against the contact, the composite in the proximal box is cured.

It has been suggested that it is not appropriate to restore both the mesial and distal surfaces at the same time, due to the difficulty in obtaining adequate proximal contacts. There is concern that it is not possible to achieve adequate interproximal separation for both proximal surfaces simultaneously. However, research has shown that better contacts are obtained when sectional matrix systems with rings are used simultaneously on the mesial and distal aspects (Fig 11-40) rather than sequentially.[261]



Fig 11-40 Simultaneous mesial and distal placement of sectional matrices and rings for an MOD restoration.

Final increment

Careful control of the final increment will minimize the amount of finishing. A number of techniques are helpful in accomplishing this goal. A rounded, cone-shaped instrument (eg, PKT3), slightly moistened with resin adhesive or a low-viscosity resin specifically designed to prevent sticking of resin composite to the instrument, may be used to shape and form the occlusal surface before curing (Fig 11-41a; see also Fig 11-14g). It is important that only a thin layer of low-viscosity resin be applied to the instrument to act as a lubricant. The best way to ensure this is

to place a drop of resin in a gauze sponge and then wipe the end of the instrument with the gauze. A fine-bristled brush (eg, sable) can be very helpful in smoothing the composite surface and achieving intimate adaptation of the resin composite to the cavosurface margins (Fig 11-41b) . Figure 11-42 shows a case of a relatively conservative tooth preparation being filled with warm composite.

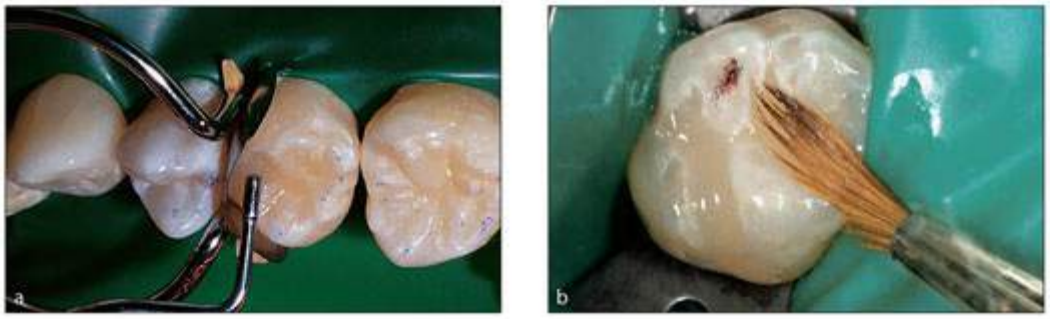


Fig 11-41*(a)* A small condenser or burnisher, lightly moistened with adhesive, is used to establish preliminary occlusal contours in the final resin composite increment before light curing. *(b)* A sable brush is used to smooth the composite surface and ensure intimate adaptation of the restorative composite to the cavosurface margins.



Fig 11-42 *(a)* Extensively demineralized and carious occlusal fissures and cavitated lesion in the mesial of the maxillary second premolar. *(b)* Proximal preparation is initiated with a no. ¼ round bur. *(c)* Initial proximal preparation, leaving a thin wall of enamel. *(d)* The thin wall of proximal enamel is removed with a spoon excavator. *(e)* Occlusal surface preparation initiated. *(f)* A no. 2 low-speed round bur is used for caries excavation. *(g)* Caries excavation completed. *(h)* A no. 7404 finishing bur is used to remove unsupported enamel and to smooth and blend external cavosurface margins. No bevels are placed on occlusal margins. *(i)* A gingival margin trimmer is used to place a gingival margin bevel. Note that the wedge has been removed to allow access for instrumentation. *(j)* Completed preparation. *(k)* An RMGI liner is placed on the dentin with a Dycal placement instrument. *(l)* Phosphoric acid etchant is placed as part of the three-step etch-and-rinse adhesive system placement. Note that a Mylar strip has been placed to protect the adjacent tooth from the etchant. *(m)* Following placement of the adhesive system, a sectional matrix band, wedge, and ring have been placed (Triodent). *(n)* The initial increment of warm composite is injected into the gingival floor of the preparation. This increment is limited to 1 mm or less in thickness. *(o)* The surface of the increment is gently smoothed with a round-end condenser

lightly lubricated with uncured resin from the adhesive system.

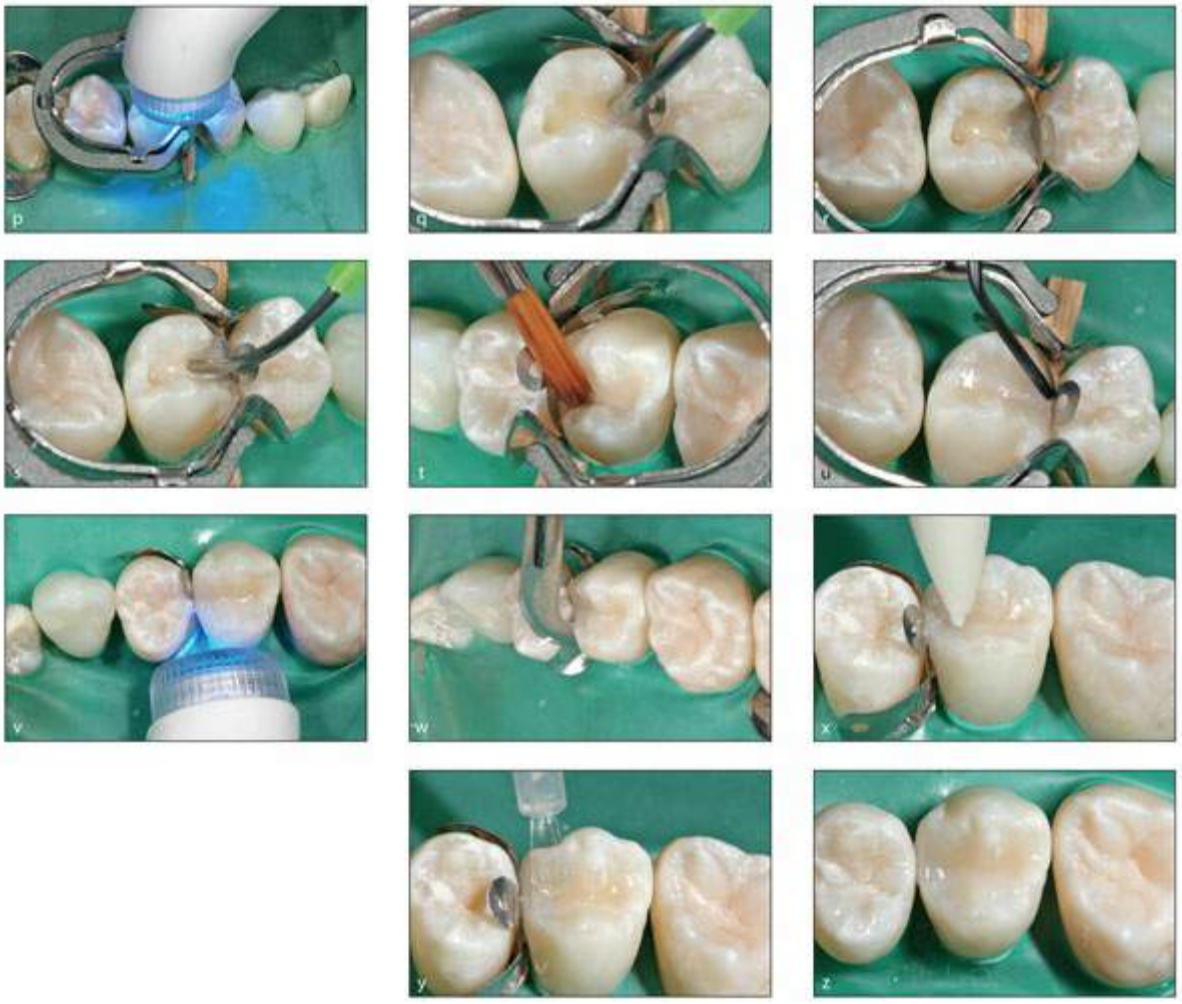


Fig 11-42 *(cont) (p)* The composite increment is cured with a light-emitting diode (LED) curing unit for 20 seconds. *(q)* The remaining increments will be placed using an oblique geometry, such that both the buccal and lingual cavity walls are not connected by the composite. Here composite is being injected into the lingual-proximal corner of the proximal box. *(r)* The proximal box has been filled to approximately the level of the axial-pulpal line angle with oblique increments of composite. At this point, remaining oblique increments can extend across the entire mesiodistal dimension of the preparation. *(s)* An oblique increment that extends across the entire mesiodistal dimension of the preparation is injected into the lingual-pulpal angle. *(t)* When the final occlusal oblique increments are placed, they should be smoothed with a fine-bristle brush, drawing the composite across the occlusal cavosurface margin to ensure intimate adaptation. *(u)* Prior to curing of the final occlusal increments, a thin, bladed instrument (eg, an IPC) is used to develop the occlusal embrasure anatomy. *(v)* Following placement of all composite increments, the wedge and ring are removed. The sectional matrix is wrapped around the proximal surface of the adjacent tooth but left in place. The restoration is then cured from the facial and lingual aspects to ensure the complete composite cure. *(w)* Finishing is initiated with a no.12B surgical blade to remove flash and initiate contour modification. *(x)* Further finishing, contouring, and polishing is accomplished with rotary instruments. *(y)* Following etching of the occlusal surface and all accessible margins, a low-viscosity rebonding agent is applied and cured. *(z)* The completed restoration.

A method for replacing occlusal anatomy and reducing finishing is called the *successive cusp build-up technique*.[322] With this procedure, incremental resin composite placement is accomplished as described in the preceding sections. However, the clinician stops the oblique layer placement and curing at a point judged to be the base of the pit and fissure anatomy for the definitive restoration. The final increments of resin composite are positioned and adapted to replace the missing portions of the inner inclines of the cusps, one cusp at a time. Because of their stiffer viscosity, packable resin composites work well in this situation. The packable resin composite can be adapted and shaped without slumping prior to curing. As each cusp is replaced with resin composite, it is briefly cured (5 seconds) to set the material in place. It is not necessary to fully cure each increment at this point, because the entire occlusal surface, and therefore all preceding increments, will be irradiated after the final cuspal incline is replaced with resin composite and is irradiated for the full curing time. This technique has been shown to provide enhanced occlusal anatomy, thereby reducing subsequent finishing. Figure 11-43 demonstrates this technique.



Fig 11-43 The successive cusp build-up technique. *(a)* Preoperative view of maxillary left molars with occlusal caries. Note that the occlusal contacts have been marked prior to rubber dam placement. *(b)* The appropriate resin composite shade is selected and transferred to a light-protected Centrix syringe tip. *(c)* The rubber plug is placed into the Centrix syringe tip. *(d)* The Centrix syringe tip is placed into the placement syringe and transferred to the composite warmer. *(e)* The completed preparations are etched. *(f)* The adhesive system is applied according to the manufacturer's instructions. *(g)* The warmed resin composite is injected into the preparations and cured in increments no thicker than 2 mm. *(h)* The initial warmed composite placement and curing is stopped at the anticipated level of the pits and fissures in the definitive restoration. Those aspects of the restoration that require little contouring for the final anatomy have been entirely filled with the warm composite (mesial/ central portions of the maxillary left first molar). *(i)* An increment of restorative composite (in this case, packable resin composite) is adapted to the lingual incline of the distofacial cusp of the maxillary left first molar. *(j)* A small instrument (PKT1) is used to form the missing cuspal anatomy in resin composite. The remaining tooth structure is used as a guide to replace the cusp inclines to the level of the fissure. The facial cuspal incline of the distolingual cusp has already been completed. *(k)* The lingual incline of the mesiofacial cusp of the maxillary left second molar is formed in resin composite to the level of the fissure. This increment is briefly cured (5 seconds). *(l)* The lingual incline of the mesiofacial cusp has been replaced in resin composite. *(m)* Each remaining cusp will be restored successively. Here the lingual incline of the distofacial cusp is being shaped with the PKT1. This increment will be cured for 5 seconds.

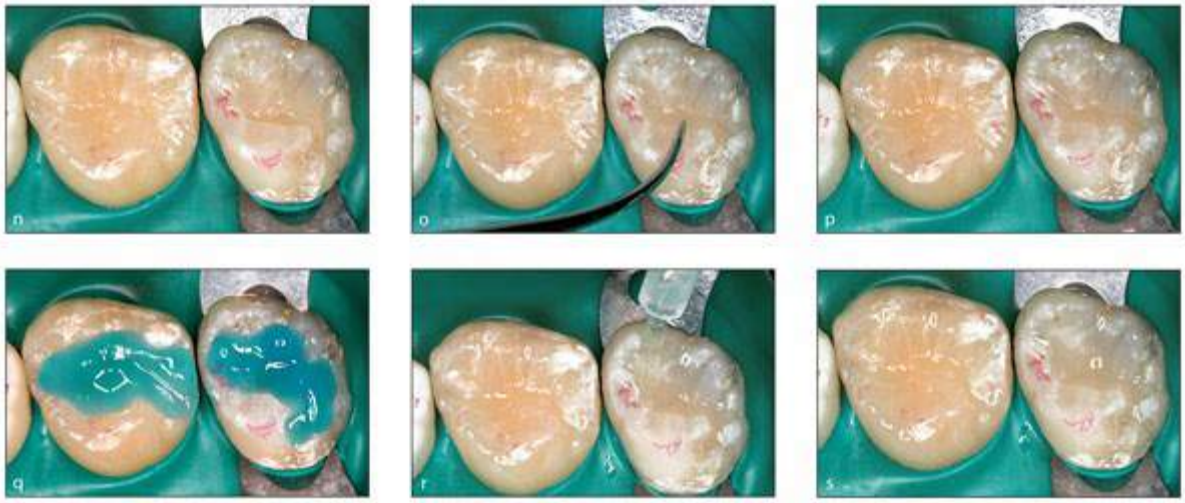


Fig 11-43 *(cont) (n)* The completed lingual inclines of the facial cusps of the second molar. *(o)* The facial incline of the lingual cusp is replaced in resin composite. Because this is the final resin composite increment, the entire occlusal surface will be light cured for the full irradition period. *(p)* The completed occlusal anatomy using the successive cusp build-up technique. While this technique takes slightly longer during placement, it reduces finishing time. *(q)* The restorations and adjacent enamel are re-etched. *(r)* The restorations are sealed with a surface sealing resin (Fortify, Bisco). *(s)* The completed restorations using the successive cusp build-up technique.

As a final step in resin composite placement and curing for Class 2 restorations, the wedge is removed and the matrix is wrapped against the adjacent tooth (Fig 11-44a). This allows access for the resin composite to be cured from the facial and lingual aspects to help ensure adequate polymerization throughout the entire restoration (see Fig 11-35). The matrix is allowed to remain in place to provide protection of the adjacent tooth during proximal-surface finishing (Fig 11-44b) and re-etching prior to sealing (Fig 11-44c).



Fig 11-44 *(a)* After the final resin composite increment has been placed and cured from the occlusal aspect, the wedge is removed and the matrix is wrapped against the adjacent tooth. This provides access for the proximal portion of the Class 2 restoration to be cured from the facial and lingual aspects to ensure the maximal cure. *(b)* The matrix is allowed to remain in place during finishing to protect the adjacent tooth from inadvertent damage when finishing the proximal portion of the restoration. *(c)* The matrix should also remain in place during re-etching and sealing to protect the adjacent tooth from being etched and bonded to the completed restoration.

Resin composite placement: Other techniques

Materials other than VLC resin composites have been suggested for placement in the proximal box preparation. Autocured (or self-curing) resin composites are available for this purpose in both low- and high-viscosity formulations. It has been suggested that the portion of the autocured resin composite adjacent to the cavity-preparation walls cures first, especially if used in conjunction with an autocured (or dual-cured) adhesive. The shrinkage of the autocured composite is said to be directed toward the cavity walls because of initiation of the curing reaction by the ongoing polymerization of the adhesive resin, and the reaction is said to be accelerated in that area by the higher temperature of the preparation walls due to body heat. This supposedly results in less polymerization shrinkage stress at the cavity margins,[158,159] thus improving marginal adaptation. However, research has failed to demonstrate either enhanced marginal adaptation[323] or reduced marginal leakage when this technique has been compared with VLC incremental placement techniques.[228]

Resin composite placement: Alternative polymerization techniques

A number of alternative techniques and devices have been introduced for curing VLC resin composites in recent years, including halogen units with variable curing intensities, lasers, plasma arc high-intensity units, and light-emitting diode (LED) curing units (Fig 11-45).



Fig 11-45*(a)* Argon laser light-curing unit (Laser-Med). *(b)* Variable-intensity halogen light-curing unit (Bisco). *(c)* Plasma arc curing high-intensity light-curing unit (DMD). *(d)* LED light-curing unit (Ultradent).

Argon laser units have demonstrated the ability to produce an increased degree of conversion of resin composite compared with standard halogen or tungsten-quartz-halogen (QTH) light-curing units.[324] In addition, the depth of cure with the laser is improved,[324] and bond strength is less affected as the light guide is moved further from the surface of the resin composite.[325] However, when the light guide is kept approximated to the composite increment, and increment depth is limited to 1.5 to 2.0 mm, no difference in bond strength is seen between that achieved with laser and standard halogen lights.[325,326] Laser light is monochromatic,[325] with the bandwidth of the laser being much narrower than that of the halogen light and centered at approximately 470 nm, which is the maximal absorption wavelength for camphorquinone, the photoinitiator most commonly used in resin adhesives and resin composites.[324,325] However, some manufacturers are starting to use proprietary photoinitiators with absorption spectra differing from that of camphorquinone, making it possible that the argon laser–curing unit would be less likely to initiate the polymerization reaction than would a halogen unit. The high price of these units, compared with other resin composite polymerization options, has limited their use in the profession.

Plasma arc curing (PAC) units generate notably higher irradiance levels than do standard halogen units. PAC units also cure resin composite at a faster rate than do the halogen curing lights.[327] Considerable evidence has been accumulated to show that this increased rate of cure does not enhance adhesion of resin composite to cavity walls. Class 5 restorations cured with argon lasers or PAC units showed significantly increased microleakage[328] and poorer marginal adaptation[329] when compared with similar restorations cured with a standard halogen curing light. This is likely due to the fact that the rate at which the modulus of elasticity, or stiffness, of the setting composite develops has a significant impact on marginal integrity. Decreasing the polymerization reaction rate allows additional time for molecular conformational changes and material flow that can relieve polymerization shrinkage stress.[78,79] This has led to research in which the polymerization reaction rate is slowed even further by reducing the irradiance of halogen curing units. Lowering irradiance to 250 mW/cm^2 has been shown to significantly improve marginal adaptation in cavity preparations versus irradiating the resin composite in those same preparations at either 450 mW/cm^2 or 650 mW/cm^2.[80,330] There has been some concern, however, that simply reducing the irradiance to these levels, while enhancing marginal adaptation, might adversely affect mechanical properties.[330] This has led to the "two-step," "soft-start," or "ramped" curing technique. Regardless of

the name, the underlying principle is the same: initial curing at a diminished irradiance to initiate the polymerization reaction at a slower rate to minimize polymerization stress, followed by a period of higher irradiance to maximize the degree of conversion and mechanical properties. In laboratory studies, this technique has proven to significantly enhance marginal adaptation without impairing mechanical properties.[82,83,329] However, research has demonstrated that resin composite factors such as shade, translucency, photoini-tiator concentration, and elastic modulus are more important in determining cure than a particular curing mode[85,331] and that curing regimens that reduce polymerization shrinkage stress to a point that it is clinically significant are not feasible.[85]

Most recently developed are the LED curing units. These units have a number of advantages compared with other curing units, including a wavelength spectrum emission that is closely matched to camphorquinone. In addition, these units are more energy efficient, allowing them to be battery operated. The diodes have a life span that is approximately 1,000 times longer than the typical halogen bulb. While the earlier versions of LED curing units provided inadequate irradiance, the newer generation has overcome this deficiency. A disadvantage to standard units is their narrow wavelength spectrum, limiting their usefulness in curing any materials that do not use camphorquinone as the photoinitiator.[332,333] However, a number of newer LED products contain chips that emit light that initiates curing of multiple photoinitiators.[334]

A 1999 evaluation of dental practices reported that 46% of the halogen visible light–curing units used in private practices provided inadequate output to cure resin composite.[335] A later study showed an improvement in that percentage, but a large percentage of halogen units still had inadequate output.[336] More recently, an evaluation of light-curing units revealed that 67.5% of halogen and 15.6% of LED light units were considered unsatisfactory.[337] It is important for the practitioner to ensure that the light-curing unit is in proper working condition and provides adequate and accurate irradiance. Testing of units can be conveniently accomplished with a radiometer (Fig 11-46).



Fig 11-46 Commercial radiometers available to monitor adequacy of irradiance of visible light–curing units. (Courtesy of USAF Dental Evaluation and Consultation Service.)

Packable resin composites

Condensable or packable resin composites were specifically designed for posterior use. In these materials, the filler type, size, and/or particle distribution differ from those of conventional restorative hybrid and microfilled resin composite materials[338,339] in order to increase viscosity and impart a consistency that more closely mimics that of dental amalgam. Claims of enhanced clinical performance, reduced polymerization shrinkage, and enhanced wear characteristics have been made. The term *condensable* is not appropriate for these materials, because condensation, by definition, denotes an increase in density, as occurs when dental amalgam is condensed into a cavity preparation. Such a volume reduction does not occur with resin composites when they are packed into a preparation.[338]

Considerable in vitro research has been accomplished to test these materials. In general, properties such as wear,[338,340–343] flexural modulus,[344,345] flexural strength,[344–346] fracture toughness,[344,347] and polymerization shrinkage$_{338}$ of the packable resin composites are comparable, but not superior, to those of other hybrid or reinforced microfilled resin composites currently available. One clinical study of a packable composite showed extreme and unacceptable wear.[348] Of particular concern with the packable resin composites was the claim by some manufacturers that these materials could be bulk-cured in thicknesses of 5 mm or greater. Independent research has clearly demonstrated that this is not the case, and adequate polymerization for most resin composites can be accomplished only in thicknesses of 2 mm or less.[214,347]

In general, practitioners can anticipate that the handling characteristics of the packable resin composite formulations may vary somewhat from other hybrid or microfilled materials. In particular, they may have a heavier consistency with a "drier" feel. Clinical trials of packable composites, mostly short-term, have

demonstrated that, with proper technique, the clinical performance of these materials can be comparable with that of other resin composites.[349–352]

The decision to use these materials should be based on individual operator preference concerning handling characteristics and not on expectations of improved clinical performance.

Finishing

Placement procedures that minimize the need for finishing and polishing should be used. The smoothest surface that can be obtained is that of unfinished resin composite that has been cured against a smooth matrix.[353–356] Finishing and polishing procedures are inherently destructive to the restoration surface and may result in the formation of microcracks at and below the surface.[357,358] Because cracks may also be produced or exacerbated during mastication, the fracture toughness of the resin composite may be significantly reduced by destructive finishing techniques.[115]

A variation of the previously described soft-start technique for posterior resin composite restorations is the *pulse delay technique*. With this technique, the final composite increment is cured for a brief period (3 to 5 seconds) at very low irradiance (150 mW/cm^2) to initiate the curing reaction at a reduced rate. After 3 to 5 minutes, the composite is again cured with a high level of irradiance. During the interim between the two curing periods, the occlusal surface is shaped and finished.[359] It should be noted, however, that the effects of manipulating this incompletely cured resin composite on physical and mechanical properties and clinical performance have not been determined by independent research.

The finishing and polishing process for posterior resin composite restorations is similar to that used with other composite restorations. A no. 12 or 12B scalpel blade, sharp no. 14L carver, Wedelstaedt chisel, or other thin, sharp-edged hand instrument is useful for removing flash from the proximal and gingival margins and for shaping proximal surfaces of resin composite (Fig 11-47). The composite material can then be finished and blended to the tooth with successively finer grits of polishing points, cups, or disks. While the use of these rotary instruments will impact the surface contours and smoothness, the long-term surface finish is more dependent on the resin composite particle size than the particular polishing system used.[360] Ideally, the finishing process used should result in a surface that has an average roughness value of less than 0.2 μm, referred to as the *critical surface finish*. The critical surface finish is the smoothness of a resin composite at which

bacterial adhesion is significantly reduced compared with a rougher surface.[361] A number of polishing systems currently on the market are able to achieve this level of polish, including Sof-Lex disc series (3M ESPE) and Astropol (Ivoclar Vivadent) and Pogo (Dentsply/Caulk) disks, cups, and points.[361,362]

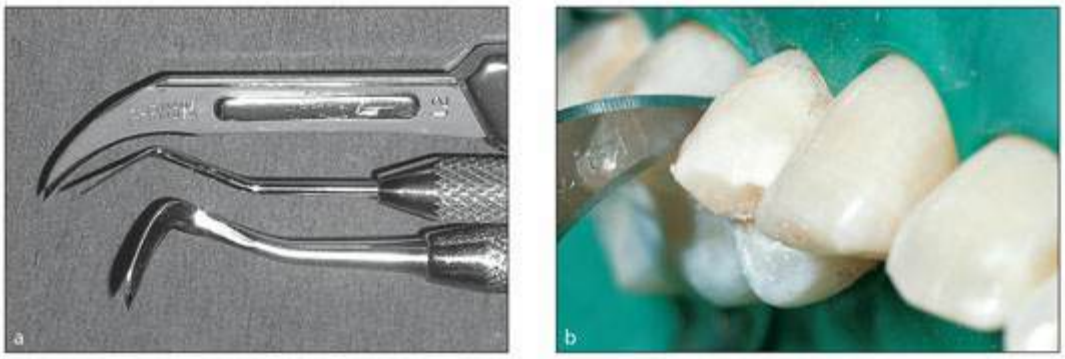


Fig 11-47 *(a)* Instruments useful in initial contouring and removal of flash: *(top)* no. 12B scalpel blade; *(middle)* IPC; *(bottom)* no. 14L carver. *(b)* No. 12B scalpel blade used to remove flash from a resin composite restoration in the distal aspect of a maxillary first premolar.

Aluminum oxide disks, used in series from coarse to very fine, tend to render some of the smoothest finishes to resin composite.[354,355,363] These work well for restoration contours that are relatively flat or convex, such as those in the facial and lingual proximal embrasure areas (Fig 11-48).



Fig 11-48 Aluminum oxide disk used to contour and polish the proximal surface of a resin composite restoration.

Abrasive disks are not practical for finishing occlusal surfaces. Shaping of these surfaces may be accomplished with multifluted carbide finishing burs or fine diamonds (Figs 11-49 and 11-50a). There is some controversy as to which of these instruments provides the smoothest surface and/or minimizes trauma-induced microleakage. Studies indicate that carbide finishing burs perform better,[363] finishing diamonds perform better,[364,365] or both perform equally well.[355,366] It is clear that the use of burs with 18 or fewer flutes leaves a significantly roughened surface, so these

burs should not be the final rotary instruments used in the finishing process.[355] Rubber or silicone disks, points, cups, and brushes impregnated with aluminum oxide, silicon dioxide, or diamond particles have been found to provide very acceptable results[353–355,360,362] and can be used to smooth the resin composite surface after initial finishing (Fig 11-50b). Finishing strips coated with aluminum oxide particles can be used to finish proximal surfaces (Fig 11-50c). As with the disks, these strips should be used in series, from coarse to very fine grit. A final high polish may be accomplished using a rubber prophylaxis cup with aluminum oxide or diamond polishing pastes.

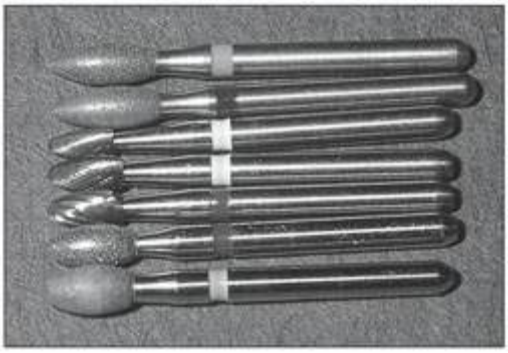


Fig 11-49 Fine diamonds and multifluted carbide burs for finishing resin composite restorations come in a variety of sizes, shapes, and grits.



Fig 11-50*(a)* Finishing diamond used to refine occlusal anatomy in a resin composite restoration. *(b)* A flexible point impregnated with aluminum oxide is used for smoothing the occlusal surface of the restoration. (*c*) Aluminum oxide finishing strip for contouring/finishing/polishing the proximal surface gingival to the interproximal contact.

Rebonding and final cure

As previously mentioned, finishing procedures are destructive to the resin composite restoration and have been shown to adversely affect wear.[207] Finishing procedures can also exacerbate the marginal gaps formed during polymerization.[66–68]

For these reasons, the practitioner should consider rebonding the occlusal surface and all accessible restoration margins with an unfilled or lightly filled VLC resin. The lower the viscosity of the rebonding resin, the more effective it will be in

penetrating interfacial gaps and microcracks.[367,368] Several low-viscosity resins, called *surface sealers*, are available for use in rebonding. While not a widely used procedure, *rebonding*, the application of a low-viscosity resin to the finished surface and margins of a restoration, has been shown to improve the marginal integrity of resin composite restorations in vitro[369] and in vivo,[210] significantly reduce microleakage in vitro,[368,370–372] and reduce marginal staining in vivo.[31] Rebonding has been demonstrated in clinical studies to significantly reduce wear and prolong marginal integrity.[31,210,369]

Although the need for etching before rebonding is somewhat controversial, phosphoric acid is usually applied to the marginal areas for 10 seconds (Fig 11-51a) and then rinsed off and the area thoroughly dried. The rebonding resin is placed, thinned with a blotted brush (Fig 11-51b) or applicator, and light cured for 20 to 40 seconds. This not only will polymerize the rebonding resin, but it may also provide additional polymerization of the resin composite.[109,135,137] To prevent the rebonding resin from joining the restored tooth to the adjacent tooth, a piece of matrix or other thin material may be placed interproximally prior to performing the rebonding procedure. Alternatively, floss is passed through the interproximal contact after the rebonding resin has been applied and before it is cured. After curing, any ledges of excess rebonding resin should be removed with a sharp-bladed instrument.



Fig 11-51 *(a)* Etching the restoration margins prior to rebonding. Note the thin plastic shim placed interproximally to protect the adjacent tooth. *(b)* Rebonding resin is brushed onto the restoration surface and margins.

The proximal contact and contours are verified with dental floss (Fig 11-52). The rubber dam is removed, and the occlusion is checked. If further occlusal adjustment is required, rebonding resin should be reapplied in the areas that were adjusted.



Fig 11-52 Proximal contact and contours are verified with dental floss. Floss is wrapped around the proximal contact to confirm the appropriate contact area.

The Tunnel Restoration

An alternative to the traditional approach for gaining access to proximal carious dentin has been termed the *tunnel preparation* (Fig 11-53). It was first suggested by Jinks[373] in 1963 as a method for placing a silver alloy mixed with sodium silicofluoride in the distal aspect of primary second molars to "inoculate" permanent first molars with fluoride as they erupted. Hunt[374] and Knight[375] later modified this procedure for use as a conservative technique for restoring teeth with small proximal caries lesions.

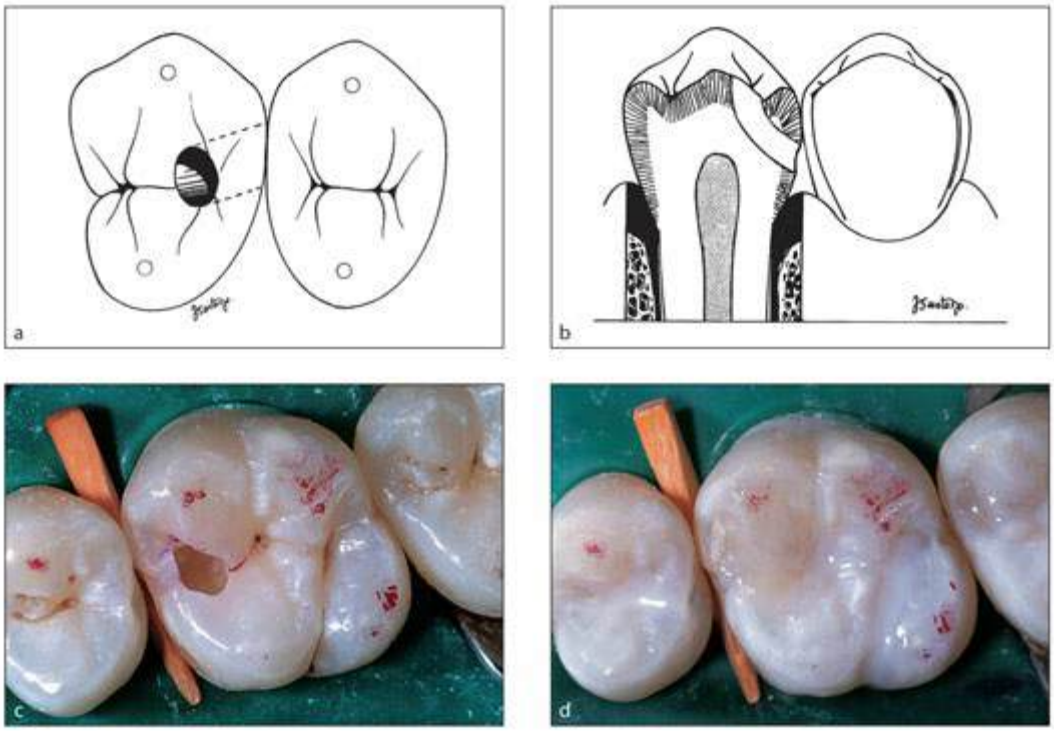


Fig 11-53 *(a)* In the tunnel preparation, access is made in the occlusal fossa adjacent to the marginal ridge. *(b)* A "tunnel" is made under the marginal ridge to the carious dentin, usually just below the interproximal contact. *(c)* Tunnel preparation access opening in the maxillary first molar. *(d)* Tunnel restoration. Occlusal fissures have been sealed.

Advantages

Several advantages are claimed for this technique. The outer surface of the proximal enamel is removed only if cavitated by caries, so there is less potential for a restorative overhang. Overhangs have been shown to occur 25% to 76% of the time with traditional Class 2 restorations; this can lead to inflammation within

periodontal tissues and bone loss.[376] With an occlusal approach, the marginal ridge is preserved, and destruction of tooth structure is minimized. A two-surface, Class 2 cavity preparation has been shown to reduce tooth stiffness by 46%; only a 20% reduction occurs with an occlusal preparation.[377] The perimeter of the restoration is reduced, decreasing the potential for microleakage.[378] Because minimal preparation is required interproximally, the potential for disturbance of the adjacent tooth is reduced. If the carious tooth structure is more extensive than originally thought and greater access is required, the preparation can easily be extended and converted into a more traditional Class 2 design.[379]

Disadvantages

Despite seemingly attractive benefits of the tunnel procedure, it remains largely unused in practice. It is a difficult procedure, demanding careful control of the preparation by the operator. The angulation of the bur for the preparation causes it to pass near the pulp. Studies have shown that the tunnel preparation often invades to within 1 mm of the pulp. A more traditional Class 2 preparation, in which penetration toward the pulp is determined by the depth of the caries lesion, tends to leave greater remaining dentinal thickness between the preparation and the pulp.[380,381] Because of the small entrance to the tunnel preparation, visibility is decreased, and removal of carious dentin is more uncertain.[374,379,382] In vitro studies on the effectiveness of caries lesion removal in the tunnel preparation have shown that there is a high rate of residual carious tooth structure after completion of the preparation.[381,383,384] For this reason, a caries-detecting solution should be used to disclose remaining carious tooth structure.

There is also concern that the marginal ridge is undermined and its strength reduced. As the diameter of the preparation increases, the marginal ridge strength decreases.[53,385,386] Although use of an adhesive restorative material has been shown to restore much of the strength of the marginal ridge,[53,385,386] this is not always the case,[380] and the degree to which marginal ridge strength is restored can depend on the size of the preparation.[385]

A number of clinical studies of tunnel restorations have been published (Table 11-1). Tunnel restorations tend to fail at a considerably higher rate than do other types of posterior restorations. Other problems noted in these studies include the fact that 34% to 41% of the restorations show either residual carious dentin or progression of enamel caries.[387,388] The most common causes of restoration failure

are marginal ridge fracture and secondary caries. In an evaluation of tunnel restorations compared with slot amalgam restorations, 21% of tunnel restorations failed during a 7-year period compared with zero failures for the slot restorations.[389] Another clinical study compared tunnel restorations in which the dentin was replaced with resin-modified glass ionomer and the remainder with resin composite to saucer-shaped Class 2 resin composite restorations over periods up to 79 months. The survival proportion for the tunnel restorations was 46% compared with 76% for the Class 2 resin composite restorations. The main cause of failure of the tunnel restorations was marginal ridge fracture, and secondary caries was significantly higher in the tunnel restorations (41% after 24 months) versus the Class 2 composite restorations (19% after 24 months).[391]

Table 11-1	**Longevity of tunnel restorations**		
Investigator(s)	**Study time (yrs)**	**No. of restorations**	**Annual failure rate**
Pilebro et al[387]	3	262	6.7
Strand et al[388]	3	161	10.0
Lumley and Fisher[389]	7	33	3.0
Hasselrot[390]	7	282	7.1
Hörsted-Bindslev et al[391]	4	54	13.4

Indications and contraindications

It is obvious from the previous discussion that tunnel restorations are not indicated and should be avoided. Other techniques discussed in this chapter (eg, the use of slot preparations) are more conservative and more predictable.

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